CHAPTER 6

Posttraumatic Stress Disorder

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The adverse psychological effects of being exposed to a traumatic event have been documented for many years. Earlier conceptualizations of traumatic stress typically regarded these reactions as transient responses that would normally abate shortly after trauma exposure. In the first edition of the *Diagnostic and Statistical Manual of Mental Disorders* (DSM; American Psychiatric Association, 1952), traumatic stress reactions were classified as acute posttrauma responses under gross stress reaction, and longer lasting reactions were subsumed under the anxiety or depressive neuroses. A major change occurred in the third edition of the DSM (American Psychiatric Association, 1980), in which the diagnosis of posttraumatic stress disorder (PTSD) was first introduced. This formal recognition of the syndrome was partly influenced by the need to understand and meet the needs of veterans returning from Vietnam with PTSD reactions. Our goal in this chapter is to provide an overview of the current state of research and clinical guidelines in acute and posttraumatic stress disorders. In the following sections, we first describe typical symptoms, course, classification, associated clinical features, epidemiology, comorbidity, etiology, and risk factors. Then we review PTSD treatment, clinical implications, and principles of change in the treatment of PTSD. We address clinical considerations throughout the text to aid in the translation of basic research into clinical practice.

TYPICAL SYMPTOMS

Posttraumatic stress disorder describes severe and persistent stress reactions after exposure to a traumatic event (American Psychiatric Association, 2013). How one

defines "trauma" varies, and the fifth edition of the DSM has specific criteria for what constitutes trauma exposure (see pp. 271–274 of DSM-5 for a description of trauma and full diagnostic criteria). PTSD comprises four symptom clusters reflecting intrusive symptoms, avoidance symptoms, negative alterations in mood and cognitions, and alterations in arousal and reactivity. Criterion D represents a new component of the PTSD diagnosis with the release of DSM-5 in that it separates avoidance from other cognitive and emotional facets of the condition. This revision was made based on accumulating evidence that effortful avoidance was distinct from mood and numbing symptoms (Elhai & Palmieri, 2011). Cognitive distortions are also a central part of PTSD and symptoms that were added to the negative alterations in cognitions and mood cluster in DSM-5 now better capture trauma-related cognitive symptoms, including negative beliefs about oneself and the world and distorted blame of self or others related to the event (LoSavio, Dillon, & Resick, 2017).

Acute stress disorder (ASD), a diagnosis first included in DSM-IV, was introduced for two primary reasons: to describe acute stress reactions that occur in the initial month after trauma exposure, and to identify trauma survivors who are high risk for developing subsequent PTSD (Harvey & Bryant, 2002). A major rationale for the introduction of this diagnosis was because PTSD can only be diagnosed at least 1 month following trauma, and there was a diagnostic gap in the initial month of trauma. The lack of a formal diagnosis to describe posttraumatic stress in the initial month potentially limited some trauma survivors' ready access to mental health services and, in this sense, a formal diagnosis was intended to alleviate this potential barrier to care. In part due to the fact that ASD is not a clear predictor of who develops PTSD, the disorder was reconceptualized in DSM-5 to broaden the required symptoms and focus on individuals with severe stress reactions in need of immediate clinical attention (Bryant, 2018). In DSM-5, the stressor criterion (Criterion A) for acute stress disorder was also changed from DSM-IV. The criterion now requires, as with PTSD, being explicit as to whether the qualifying traumatic events were experienced directly, witnessed, or experienced indirectly. Also, DSM-IV Criterion A2 regarding the subjective reaction to the traumatic event (e.g., "the person's response involved intense fear, helplessness, or horror") was eliminated. Based on evidence that acute posttraumatic reactions are very heterogeneous and that DSM-IV's emphasis on dissociative symptoms was too restrictive, individuals may meet diagnostic criteria in DSM-5 for acute stress disorder if they report any nine of 14 listed symptoms in these categories: intrusion, negative mood, dissociation, avoidance, and arousal. Support for the inclusion (and previous focus on DSM-IV) of dissociative symptoms in the ASD diagnosis to predict subsequent PTSD came from evidence demonstrating an association between "peritraumatic dissociation" (i.e., dissociation occurring in the immediate aftermath of a traumatic experience) and subsequent levels of PTSD, a finding that has been replicated across numerous longitudinal studies (Murray, Ehlers, & Mayou, 2002).

COURSE

There is strong evidence that most people recently exposed to a traumatic experience commonly report posttraumatic stress reactions in the initial weeks after trauma.

Despite the prevalence of acute stress reactions, the majority of these stress responses are transient (Bryant, 2003; Galatzer-Levy, Huang, & Bonanno, 2018). For example, whereas 94% of rape victims displayed sufficient PTSD symptoms 2 weeks posttrauma to meet criteria (excluding the 1-month time requirement), 11 weeks later this rate dropped to 47% (Rothbaum, Foa, Riggs, Murdock, & Walsh, 1992). In another study, 70% of women and 50% of men were diagnosed with PTSD at an average of 19 days after an assault; the rate of PTSD at 4-month follow-up dropped to 21% for women and zero for men (Riggs, Rothbaum, & Foa, 1995). Similar patterns have been observed following motor vehicle accidents (Blanchard, Hickling, Barton, & Taylor, 1996), the New York terrorist attacks (Galea et al., 2003), and the 2004 tsunami (van Griensven et al., 2006).

Despite a general trend for people to adapt after trauma, it is also important to note that there are different trajectories from acute reactions to chronic PTSD. The main exemplar of a different trajectory is delayed-onset PTSD. Delayed onset is an uncommon response in civilian populations, although there is some evidence that it is more common in military contexts (for a review, see Andrews, Brewin, Philpott, & Stewart, 2007). Although the mechanisms underpinning delayed-onset PTSD are not well understood, several explanations have been offered. These include the implementation of numbing responses that inhibit expression of distress (Horowitz & Solomon, 1975), preoccupation with more immediate needs (e.g., pain, surgery) that distract attention from one's symptoms (Andreasen, 2004), the increase of stressors in the posttrauma period that compound the initial stress response (Bryant & Harvey, 2002), or rising demands on cognitive resources that are required to manage emotional responses and then lead to increased distress (Grossman, Levin, Katzen, & Lechner, 2004). There is evidence that it represents subsyndromal levels of PTSD that accumulate over time, potentially as a function of increasing posttraumatic stress; that is, prospective evidence indicates that cases described as delayed onset actually display elevated levels of posttraumatic stress in the initial period after the trauma (Andrews et al., 2007; Horesh, Solomon, Keinan, & Ein-Dor, 2013). Interesting evidence on this issue emerged in a study on the trajectory of adaptation after Hurricane Katrina, which indicated that rates of PTSD actually increased over time following Hurricane Katrina (Kessler et al., 2008). It is possible that the lengthy periods of relocation, lack of housing, and loss of basic infrastructures resulted in increased psychological strain, which led to rising rates of PTSD. As mentioned earlier, delayed-onset PTSD appears to be more common in military settings than in civilian contexts, with accumulating evidence from military populations that soldiers do not appear to have PTSD immediately after deployment but do develop the disorder months later (Andrews et al., 2007; Horesh, Solomon, Keinan, & Ein-Dor, 2013). One possibility is that the relief of leaving the immediate context of threat and the attention given to soldiers in the immediate postdeployment environment result in a period of minimal posttraumatic stress; however, as time passes, they are increasingly aware of the negative aftereffects of their trauma.

It is also worth noting that our understanding of the course of traumatic stress has become more sophisticated in recent years, as studies have included multiple time points that allow closer analysis of the course of PTSD over time. For example, one large study of traumatic injury patients revealed that rather than following a linear course, at least half of people who have PTSD at one time may not have it at

a subsequent assessment—and then have a worsening of their condition at another subsequent assessment (Bryant, O'Donnell, Creamer, McFarlane, & Silove, 2013). This evidence highlights that PTSD can be a dynamic condition that tends to fluctuate according to current life stressors. The course of PTSD has also been the subject of many studies that have used latent growth mixture modeling, which allows identification of classes of individual variation over time and describes distinct classes of trajectory. These studies have reliably demonstrated (1) a resilient class that consistently shows few PTSD symptoms, (2) a recovery class with initial distress then gradual remission, (3) a delayed reaction class with initial low symptom levels but increased symptoms over time, and (4) a chronic distress class with consistently high PTSD levels (Bryant et al., 2015; deRoon-Cassin et al., 2010; Galatzer-Levy et al., 2018). There is also evidence using network analysis that PTSD as a cohesive construct develops in the months after trauma exposure; this statistical approach focuses on relationships between symptoms, which shows that PTSD symptoms are more loosely interconnected in the acute posttrauma phase, and become more cohesively interrelated in the following months (Bryant et al., 2017).

This convergent evidence underscores the challenges of predicting subsequent PTSD from acute reactions because PTSD does not appear to follow a linear course. Although there is evidence of an association between elevated PTSD symptoms in the acute phase and development of later PTSD (Jehel, Paterniti, Brunet, Duchet, & Guelfi, 2003; Karstoft, Galatzer-Levy, Statnikov, & Shalev, 2015; Shalev & Freedman, 2005), these reported associations do not provide us with adequate cutoffs to reliably determine who will develop PTSD. One way of improving possible early detection of who will develop PTSD comes from a consortium that recently combined 2,473 trauma survivors from 10 longitudinal studies that used a likelihood estimate approach (Shalev et al., 2019). This study revealed that if a patient reported elevated PTSD severity, female gender, less than secondary-level education, and history of interpersonal trauma, that person had a 34% greater likelihood of developing PTSD than someone without these risk factors. The type of trauma is also a critical factor to consider, with assaultive injuries being particularly predictive of subsequent development of PTSD (Lowe et al., 2020).

CLASSIFICATION

Posttraumatic Stress Disorder

In the discussions preceding DSM-5, considerable attention was devoted to the classification of PTSD, in terms of both where it belongs in the broader spectrum of psychiatric disorders and its own definition. One important decision was to reclassify PTSD in a new category of trauma-related and stressor-related disorders (rather than with the anxiety disorders, where it appeared in DSM-IV). Debate focused on the possibility of identifying a subset of anxiety disorders termed "fear circuitry disorders" that arise after a traumatic/stressful event. Some experts proposed that this category would include PTSD, panic disorder/agoraphobia, social anxiety disorder, and specific phobia (Andrews, Charney, Sirovatka, & Regier, 2009). Although there is some evidence that aversive or traumatic experiences do precede onset of panic disorder (Manfro et al., 1996) and social phobia (McCabe, Antony, Summerfeldt, Liss, & Swinson, 2003), other evidence suggests a different pattern (Rapee, Litwin, & Barlow,

1990). There is also strong evidence that a range of other disorders develop after trauma exposure. Depression is a commonly observed psychological disorder in survivors of traumatic events, along with various anxiety disorders (Bonde et al., 2016; Knowles, Sripada, Defever, & Rauch, 2019; Rytwinski, Scur, Feeny, & Youngstrom, 2013). After the Oklahoma City bombing, 22% of people suffered depression, 9% had alcohol use disorder, 7% suffered panic disorder, 4% had generalized anxiety disorder, and 2% had drug use disorder; overall, 30% of people had a psychiatric disorder other than PTSD (North et al., 1999). In another study of over 1,000 traumatic injury survivors, depression was the most common disorder (16%), followed by generalized anxiety disorder (11%), agoraphobia (9%), and PTSD (9%) (Bryant et al., 2010). These patterns highlight that PTSD is not the only condition that can develop following trauma. However, the evidence that these other disorders are not distinctive to the aftermath of trauma is not consistent with their being part of a fear circuitry category. Another challenge is that the fear circuitry construct does not give adequate attention to other emotions associated with PTSD, including sadness, grief, anger, guilt, and shame (Friedman, Resick, Bryant, & Brewin, 2011).

One line of research supporting the fear circuitry category comes from neuroimaging research. The prevailing model of fear circuitry posits that these disorders are characterized by impaired medial prefrontal cortex activation, which leads to excessive amygdala response, and this network is responsible for fear reactivity (Charney, 2003). An increasing body of evidence indicates that whereas fear circuitry disorders tend to be characterized by excessive amygdala reactivity, and to a lesser extent impaired regulation of that response by the medial prefrontal cortex (e.g., Shin & Liberzon, 2010), different neural networks appear to be involved in nonfear circuitry anxiety disorders (e.g., Duval, Javanbakht, & Liberzon, 2015; Rauch et al., 2007). It is important to acknowledge the heterogeneity of PTSD as a disorder and its components beyond fear-based symptoms. For example, in considering the anhedonic symptoms of PTSD, deficits in reward processing for individuals with PTSD have also been found (see Nawijn et al., 2015, for a review). Additionally, there is evidence for the dissociative subtype characterized by increased activation of medial prefrontal regions and hyperinhibition of the amygdala and other limbic regions (Lanius, Brand, Vermetten, Frewen, & Spiegel, 2012).

Although we have thus far focused primarily on the diagnosis of PTSD based on DSM-5, it is important to note differences between how PTSD is represented in DSM-5 and in the 11th revision of the *International Classification of Diseases and Related Health Problems* (ICD-11; World Health Organization [WHO], 2019), and how that may affect case conceptualization and treatment. As mentioned earlier, DSM-5 (American Psychiatric Association, 2013) includes 20 symptoms and four symptom clusters: intrusions, active avoidance, negative alterations in cognitions and mood, and arousal. A dissociative subtype reflecting the presence of either derealization or depersonalization is also included. The ICD-11 has a narrower definition of PTSD than does the DSM-5. Specifically, PTSD includes only six symptoms, categorized into three core symptom clusters: intrusions, avoidance of trauma-related thoughts and behaviors, and an ongoing sense of threat (hyperarousal and startle response) (Maercker et al., 2013). The symptoms must cause significant functional impairment and have been present for several weeks. A major goal of the revision to ICD-11 PTSD criteria was to reduce symptom overlap with other disorders and focus on symptoms that best discriminate PTSD from other disorders (Brewin et al., 2017). There has

been some evidence that the ICD-11 definition of PTSD better fits data across varied trauma types and reduces comorbidity with depression in comparison to the DSM-5 definition of PTSD (Hansen, Hyland, Armour, Shevlin, & Elklit, 2015; Morina, van Emmerik, Andrews, & Brewin, 2014). Other research suggests that the narrow ICD-11 definition of PTSD may result in underdiagnosis of PTSD and may miss individuals in need of treatment (Hafstad, Thoresen, Wentzel-Larsen, Maercker, & Dyb, 2017; Wisco et al., 2016).

Complex PTSD (CPTSD) is a new diagnosis in ICD-11 characterized by disturbances in self-organization that often result from exposure to chronic trauma that cannot be easily escaped (e.g., childhood sexual abuse, intimate partner violence, torture) (Maercker et al., 2013). The ICD-11 CPTSD diagnosis requires the presence of PTSD based on the three ICD-11 symptom clusters described earlier and includes three additional symptom clusters related to disturbances in self-organization, including affect dysregulation, negative self-concept, and difficulties in relationships (Cloitre, Garvert, Brewin, Bryant, & Maercker, 2013; WHO, 2019). CPTSD was first introduced by Herman (1992), who described the impact of repeated traumatic stressors on self-organization that can develop uniquely from PTSD symptoms. The ICD-11 CPTSD diagnosis stems from the DSM-IV disorder of extreme stress not otherwise specified (DESNOS) diagnosis (van der Kolk, Roth, Pelcovitz, Sunday, & Spinazzola, 2005). Numerous studies have supported CPTSD as a distinct disorder (Brewin et al., 2017; Cloitre et al., 2013). However, there remains controversy over the clinical utility of CPTSD beyond other disorders with overlapping symptoms, such as PTSD and borderline personality disorder (Jowett, Karatzias, Shevlin, & Albert, 2020; Wolf et al., 2015), and as a result, CPTSD was not included as a diagnosis in DSM-5 (Resick et al., 2012).

Acute Stress Disorder

Since its introduction, the ASD diagnosis has been widely criticized for several reasons. In terms of describing acute posttraumatic stress, approaches that focus on fear responses alone may neglect other psychological reactions (Isserlin, Zerach, & Solomon, 2008). In contrast to ASD, the construct of acute stress reactions described in ICD-11 (WHO, 2019) conceptualizes acute stress reaction as a transient reaction that can be evident immediately after the traumatic event and usually resolves within 2–3 days after trauma exposure. This transient reaction may include a range of emotional, cognitive, behavioral, or somatic reactions. Importantly, acute stress reaction is no longer included as a diagnosis in ICD-11, but instead is included in the factors influencing health section in the context of problems associated with harmful or traumatic events. This change was made in part in order to normalize reactions to trauma, while still allowing for clinical intervention as a response to amorphous acute stress reaction as needed (Maercker et al., 2013). It is important to note that ICD-11 does not hold the same 1-month posttrauma requirement for PTSD diagnosis as DSM-5, so the need to address symptoms within the first month following trauma exposure may be somewhat reduced.

A secondary goal of the ASD diagnosis is to discriminate between recent trauma survivors who are experiencing transient stress reactions and those who are suffering reactions that will persist into long-term PTSD. A large number of prospective studies have assessed trauma survivors for ASD in the initial month after trauma

and subsequently assessed for PTSD months or years later (for a review, see Bryant, 2011). Across these studies, at least half of trauma survivors with ASD subsequently meet criteria for PTSD. In contrast, the majority of survivors who eventually developed PTSD did not initially meet ASD criteria. This pattern suggests that if a major goal of ASD is to predict who will subsequently develop PTSD, it is failing to identify half of those who will meet criteria for PTSD at some later time.

DSM-5 reformulated the construct of ASD so that its basis is not solely as a state that predicts PTSD. Evidence over the past two decades suggests that ASD does not seem to be able to identify the majority of people who develop PTSD upon exposure to a traumatic event (Bryant, Friedman, Spiegel, Ursano, & Strain, 2011). For the current DSM edition, the goal is to describe severe acute stress reactions that may or may not be transient responses but might nonetheless benefit from mental health interventions. The ASD definition now requires that the patient satisfy a minimum of nine out of a total 14 potential symptoms in recognition that ASD presentations are heterogeneous.

Clinical Considerations with Classification

Since trauma is so ubiquitous and certain types of trauma exposure are related to increased risk for PTSD, it is important to assess for trauma exposure and related PTSD symptoms in clinical assessments to ensure that something critical is not missed. As described earlier, PTSD is not the only psychological disorder to develop as a result of trauma, so an individual's presenting concern may not include trauma-specific symptoms. Clinicians should keep in mind that the client's understanding of the potential connectedness between psychological symptoms and trauma exposure may vary. Additionally, the dynamic nature of PTSD means that symptoms may come and go, and may be influenced by changing environmental factors or stress. When asking about trauma exposure, it is helpful to ask about specific traumas and obtain enough information to understand broadly what happened without asking for so much detail as to unnecessarily retraumatize the individual during the assessment process. As described later, a major component of trauma-focused treatment is going into significant detail about the trauma memory; however, in the context of initial clinical assessment and case conceptualization, that level of detail is unnecessary. If a patient is sharing significant detail early on, it can be helpful to praise their openness and acknowledge that there will be later opportunities to go into detail, while gently redirecting to the goals of the assessment sessions. Exact diagnosis will clearly depend on the classification system that is used; however, it is helpful to remember that even clinically significant subthreshold symptoms that cause functional impairment can be addressed in the context of trauma-focused treatment.

ASSOCIATED CLINICAL FEATURES

There is much evidence that PTSD is linked to a wide range of other mental health and psychosocial problems, both of which contribute to impaired functioning and also compound the PTSD itself. PTSD has been linked to a large range of impairment indicators, including difficulties with employment, homelessness, social engagement, intimate partner relationships, and parenting, as well as poorer self-rated

health and higher utilization of medical services (Rodriguez, Holowka, & Marx, 2012; Tuerk et al., 2013). Impairment is associated with a wide range of PTSD symptoms (Ross, Murphy, & Armour, 2018), although there is evidence that avoidance and numbing are the PTSD symptoms that most contribute to impairment (Shea, Vujanovic, Mansfield, Sevin, & Liu, 2010). There is convergent evidence that trauma exposure is linked to a range of poor physical health outcomes, and it appears that this effect can be largely attributed to PTSD (Schnurr, Wachen, Green, & Kaltman, 2014). Several mechanisms have been proposed for this relationship, including biological (e.g., increased cardiovascular reactivity, altered hypothalamic–pituitary–adrenal [HPA] activity), behavioral (e.g., increased substance use, smoking), and psychological (e.g., chronic depression, hostility) factors that may impair health (Friedman & McEwen, 2004). PTSD is also linked to impaired sleep function (Germain, McKeon, & Campbell, 2017); not only do nightmares and hyperarousal lead to sleep disturbance, but also sleep disturbance immediately prior to trauma exposure enhances risk for PTSD (Bryant, Creamer, O'Donnell, Silove, & McFarlane, 2010). Finally, it is important to note that suicide risk is increased with PTSD across veteran and civilian populations (Gradus, 2018; Krysinska & Lester, 2010). The association between PTSD and suicide is complex, and the issue of suicide risk in PTSD is particularly relevant because of the elevated rates of suicidality in other mental disorders (e.g., depression) that are highly comorbid with PTSD.

EPIDEMIOLOGY

Population studies have shown that the majority of people in the community have been exposed to traumatic stressors. The U.S. National Comorbidity Survey indicated that 61% of randomly sampled adults reported exposure to a traumatic stressor (Kessler, Sonnega, Hughes, & Nelson, 1995). More recent national- and community-level assessments indicate that closer to 90% of adults are exposed to a traumatic stressor (Breslau, 2009; Kilpatrick et al., 2013). Despite the frequency of exposure to potentially traumatizing events, relatively few people actually develop PTSD. For example, the National Comorbidity Survey revealed that only 20.4% of the women and 8.2% of the men ever developed PTSD (Kessler et al., 1995). Similarly, based on DSM-5 criteria, the National Stressful Events Survey revealed that only 12.8% of the women and 5.7% of the men had developed PTSD (Kilpatrick et al., 2013). Consistent evidence supports greater risk for the development of PTSD in women than men that is not attributable to type or frequency of trauma exposure (Breslau, 2009); although factors increasing women's risk are not fully understood, there is growing evidence of the importance of neurobiological differences that may contribute to women's increased risk (Pineles, Hall, & Rasmusson, 2017).

Prevalence rates are not the same everywhere. Across countries and cultural settings there is marked variability in the occurrence of PTSD. The World Mental Health Survey has conducted the most global analysis of PTSD rates, and this study found that DSM-IV-defined PTSD (12-month prevalence rates) was higher in high-income countries (Northern Ireland = 3.8%; United States = 2.5%; New Zealand = 2.1%) than in low-income countries (Columbia = 0.3%; Mexico = 0.3%) (Karam et al., 2014). These studies indicate that the normative response following trauma exposure is to adapt to the experience and to not develop PTSD.

There is good evidence that PTSD risk is associated with different types of traumatic events. The risk for PTSD is lower following natural disasters (5–10%) relative to sexual assault (>40%) (Creamer, Burgess, & McFarlane, 2001; Kessler et al., 1995). Also, interpersonal violence typically is associated with higher risk for PTSD, and particularly following sexual violence (Forbes et al., 2012, 2014). The World Mental Health Survey reported that organized, physical, or sexual violence increased risk for PTSD (Liu et al., 2017). In short, a person is more likely to develop PTSD when they experience prolonged and severe trauma. This includes individuals living in civil conflict and refugees (Charlson et al., 2019). Military combat is also associated with increased risk for PTSD (Kilpatrick et al., 2013).

It is worth noting that comparing prevalence rates across studies is difficult because of methodological differences, different patterns in trauma exposure, variability in access to health care, and cultural distinctiveness in the definition of posttraumatic symptoms. Accordingly, to address the role of race and ethnicity in PTSD, it is important to consider racial and ethnic variation in similar settings, with exposure to comparable traumatic events. In this context, there are different prevalence rates across racial and ethnic groupings. For example, in the wake of the September 11, 2001, terrorist attack in New York, Hispanic, Black, and other racial and ethnic/minority individuals were more likely than White individuals to develop PTSD (Adams & Boscarino, 2005; DiGrande et al., 2008). Of note, a recent review shows evidence of increased conditional risk for PTSD in Latinx adults (Alcántara, Casement, & Lewis-Fernandez, 2013). In addition to civilian samples, there is also evidence in military samples that minorities are particularly at risk of PTSD development (Schlenger et al., 1992; Smith, Goldstein, & Grant, 2016). It is difficult to identify the reason for this pattern; it may be attributable to lower socioeconomic status and associated resource availability to manage trauma, the ongoing chronic stress of racial discrimination, or cultural differences in expressing and coping with adversity (Ruef, Litz, & Schlenger, 2000). Evidence of lifetime rates of PTSD as high as 50% in low-income urban minority communities (Gillespie et al., 2009) highlights how critical it is to understand the intersection between sociocultural factors that increase risk and ensure that interventions are developed to target those at highest risk.

Clinical Considerations for Epidemiology

There is significant awareness of the need to treat PTSD and other trauma-related symptoms in veterans, and many resources have been put toward these efforts. However, for other at-risk groups, such as low-income minority individuals, awareness and resources are more limited. While stigma and barriers exist for treatment engagement in both veterans and civilians, these are particularly present for civilians who often are unaware that PTSD is something other than a "war-related" problem. For those working in settings that serve underserved and underprivileged communities, trauma-informed care and screening for trauma and PTSD are critical.

COMORBIDITY

There is overwhelming evidence that PTSD is more often associated with other psychiatric disorders than occurring as the sole diagnosis. Lifetime comorbidity prevalence

rates with PTSD have been reported in at least two-thirds of cases (Kessler et al., 1995). Recent estimates in veterans with PTSD show similar rates (Smith et al., 2016). The major overlap is between PTSD and depression, anxiety disorders, and substance use disorders. Although it is possible that a degree of comorbidity may be attributed to overlapping symptoms between PTSD and both depressive and anxiety disorders, evidence indicates that there is marked comorbidity with these disorders and PTSD, even when researchers control for this overlap (Flory & Yehuda, 2015). There is also considerable comorbidity with personality disorders (Smith et al., 2016). PTSD has frequently been shown to be comorbid with borderline personality disorder, which arguably is not surprising, as there is often a significant trauma history in the profiles of those with borderline personality disorder (Golier et al., 2003). There is also significant overlap between PTSD and antisocial personality disorder (Sareen, Stein, Cox, & Hassard, 2004); it has been suggested that the antisocial tendencies may precede the PTSD, and that dangerous behaviors may predispose the person to suffer traumatic events.

Two major mechanisms have been proposed to explain these high rates of comorbidity. One pathway may involve people developing a psychiatric disorder, which predisposes them to experiencing trauma and associated PTSD. The other pathway is the primary onset of PTSD, which leads to simultaneous or subsequent comorbid disorders. It appears that a significant proportion of comorbid cases reflect the development of PTSD after the initial onset of another disorder (Breslau, 2009). Anxiety disorders and substance use disorders may be particularly influential in increasing risk for trauma exposure and PTSD (e.g., Stein et al., 2002). There is also strong evidence that much comorbidity development follows PTSD onset, implying that PTSD can increase the likelihood of developing distinct disorders (Breslau, 2009; Perkonigg, Kessler, Storz, & Wittchen, 2000). One important consequence of PTSD is increased substance use, which has been conceptualized as a form of self-medication and can lead to marked long-term dysfunction in the context of chronic PTSD (Hawn, Cusack, & Amstadter, 2020; Ouimette, Read, Wade, & Tirone, 2010).

ETIOLOGY

Learning and Biological Theories of PTSD

Biological models of PTSD have predominantly focused on fear conditioning concepts to explain the genesis and maintenance of PTSD (Milad, Rauch, Pitman, & Quirk, 2006). Fear conditioning models posit that when a traumatic event (unconditioned stimulus) occurs, people typically respond with fear (unconditioned response). It is argued that the strong fear elicited by the trauma will lead to strong associative conditioning between the fear and the events surrounding the trauma. As reminders of the trauma occur (conditioned stimuli), people then respond with fear reactions (conditioned response). It is postulated that release of stress neurochemicals (including norepinephrine and epinephrine) in the context of the traumatic experience results in very strong conditioning and overconsolidation of trauma memories (Rauch, Shin, & Phelps, 2006). This model proposes that successful recovery from trauma involves extinction learning, in which repeated exposure to trauma reminders or memories results in new learning that these reminders no longer signal threat (Davis & Myers, 2002).

In support of these proposals, there is evidence that people who eventually develop PTSD display elevated resting heart rates (Shalev et al., 1998) and respiration rates (Bryant, Creamer, O'Donnell, Silove, & McFarlane, 2008) in the initial days after trauma. There have been reports that other indices of elevated arousal in the immediate posttrauma phase are associated with subsequent PTSD, such as lower gamma-aminobutyric acid (GABA) levels (Vaiva et al., 2006). Psychophysiological studies during the acute trauma phase have also explored the responses of individuals using their own recollections of the trauma as the eliciting stimulus. A study of patients in the Emergency Department posttrauma found that higher skin conductance response during a bedside trauma interview was predictive of the development of chronic PTSD (Hinrichs et al., 2019). Other paradigms using trauma narratives in the acute trauma phase also find larger psychophysiological responses, including heart rate and skin conductance among those who go on to develop PTSD (Goodman & Griffin, 2018; Gutner et al., 2010). Importantly, psychophysiological response (acoustic startle response, skin conductance) is also an indicator of treatment response in trauma-focused treatment (Maples-Keller et al., 2019; Wangelin & Tuerk, 2015).

Biological models propose that PTSD occurs when the amygdala is hyperresponsive to fear-related stimuli and there is a concomitant lack of "top-down" inhibition from the medial prefrontal cortex (MPFC; Rauch et al., 2006). There is reduced MPFC recruitment in PTSD during fear processing, particularly in rostral anterior cingulate cortex (Hayes, Hayes, & Mikedis, 2012; Shin et al., 2005). There is also evidence for increased amygdala response to fear stimuli in PTSD (see review by Patel, Spreng, Shin, & Girard, 2012), although individuals with the dissociative subtype of PTSD may not show this pattern and instead show hyperinhibition of the amygdala and other limbic regions (Lanius et al., 2012). Impairment in hippocampal activation is also implicated in individuals with PTSD, representing deficits in memory and in the ability to use contextual cues appropriately in the context of fear learning (Joshi, Duval, Kubat, & Liberzon, 2020). It is worth noting that trauma-focused therapy is conceptualized as a form of extinction learning, in which conditioned fear responses are inhibited by learning of new, safe associations (Rothbaum & Davis, 2003). In this context, it is interesting to note that the same neural networks implicated in extinction learning are also involved in successful cognitive-behavioral treatment (CBT; described in the treatment section below). The neural mechanisms implicated in extinction learning (i.e., MPFC–amygdala network) have been shown to predict positive response to CBT in patients with PTSD in both functional (Bryant, Felmingham, Kemp, et al., 2008) and structural (Bryant, Felmingham, Whitford, et al., 2008) magnetic resonance imaging (MRI). Furthermore, as patients with PTSD successfully respond to CBT, they are better able to recruit the MPFC during fear processing (Felmingham et al., 2007; Shou et al., 2017).

Cognitive Models of PTSD

Cognitive models of PTSD posit that the development and maintenance of PTSD rely heavily on two processes: (1) maladaptive appraisals of the trauma and its aftermath, and (2) disturbances in autobiographical memory (Ehlers & Clark, 2000). These models posit that trauma memories are encoded in a fragmented manner because of the elevated arousal that occurs at the time of trauma. It is proposed that the

disorganized nature of these memories precludes a coherent narrative, and this can impede adaptive processing of the experience and limit the extent to which the traumatic experience can be placed in the context of other nontraumatic memories the person may retrieve. It is also proposed that the manner in which people appraise the traumatic event, their responses to it, and their environment after the trauma are pivotal in terms of perpetuating their sense of threat or sense of vulnerability.

There is a great body of evidence supporting this model. Individuals with PTSD have an attentional bias to threat stimuli (e.g., Fani et al., 2012; McNally, Kaspi, Riemann, & Zeitlin, 1990; Naim et al., 2015; Powers et al., 2019), are deficient in recalling specific positive memories (McNally, Lasko, Macklin, & Pitman, 1995; Schönfeld & Ehlers, 2017), and engage in maladaptive thoughts about how they responded to the trauma or to future negative events that may happen to them (Dekel, Peleg, & Solomon, 2013; Warda & Bryant, 1998; Zuj et al., 2017). Catastrophic appraisals about the experience shortly after the trauma are also predictive of subsequent PTSD (Ehlers, Mayou, & Bryant, 1998; Nixon & Bryant, 2005).

RISK FACTORS

The robust finding that only a minority of trauma survivors develop PTSD raises the issue about variables that increase risk for PTSD development. In terms of preexisting risk factors, certain biological and cognitive characteristics (e.g., HPA axis dysfunction, pretrauma cognitive biases), prior psychological disturbance, family history of psychological disorders, childhood maltreatment, and female gender are predictive of PTSD development (Bomyea, Risbrough, & Lang, 2012; Brewin, Andrews, & Valentine, 2000; Trickey, Siddaway, Meiser-Stedman, Serpell, & Field, 2012). Recent studies in trauma patients have also indicated that higher levels of emotion dysregulation at the time of trauma exposure increases risk for the development of PTSD independent of co-occurring psychiatric symptoms (Forbes et al., 2020; Pencea et al., 2020). In terms of factors associated with the trauma, a major predictor of PTSD is the nature of the traumatic event itself. The more one is exposed to threat and to grotesque or distressing events, the more likely one is to develop PTSD (Breslau, 2009; Brewin et al., 2000). As noted earlier in the discussion of ASD, there is considerable evidence that peritraumatic responses can predict subsequent PTSD. Symptom severity, including dissociation, reexperiencing, avoidance, and arousal, several weeks after exposure can be predictive of subsequent PTSD. Consistent with cognitive models, there is evidence that catastrophic appraisals of the stressor and of one's reactions to the stressor influence whether chronic PTSD develops (Ehlers et al., 1998). In relation to posttrauma factors, there is evidence that self-reported levels of low social support and the presence of ongoing stressors are important predictors of PTSD (Brewin et al., 2000; Nillni et al., 2013).

Most of the literature on risk factors for PTSD has been limited by inferring risk on the basis of data collected retrospectively after trauma exposure; this data can be confounded by mood biases in retrospective reports. In more recent years, researchers have focused on populations with high risk for trauma exposure and have assessed these people prior to trauma exposure, with subsequent follow-up after trauma exposure. A systematic review (DiGangi et al., 2013) of all prospective

studies from 1991 to 2013 that included measures both pre- and posttrauma exposure, found that lower cognitive ability, negative cognitive bias, personality characteristics (e.g., neuroticism, hostility), prior psychopathology and trauma exposure, higher levels of physiological arousal, negative family environment, poor social support, and poverty all increased risk for PTSD. Recent evidence with soldiers pre- and postdeployment indicates related patterns. For example, predeployment neuroticism and dissociation are associated with postdeployment development of PTSD (Ponce de León, Andersen, Karstoft, & Elklit, 2018). Lower levels of heart rate variability (HRV) predeployment predict higher levels of PTSD postdeployment (Minassian et al., 2015). Additionally, certain attention biases predeployment, namely, cognitive avoidance of threat-related stimuli, are predictive of greater PTSD symptoms postdeployment even in the context of less war zone stress (Beevers, Lee, Wells, Ellis, & Telch, 2011). There is also evidence that the tendency to engage in catastrophic thinking before trauma exposure predicts subsequent PTSD in firefighters (Bryant & Guthrie, 2005, 2007).

Twin studies indicate the role of genetic contribution, in that heritability accounts for up to 30% of the variance in PTSD symptoms (True et al., 1993). It is important to note, however, that the genetic influences on PTSD are common to other (frequently comorbid with PTSD) disorders, including depression, other anxiety disorders, and substance use (Koenen et al., 2008; Smoller, 2016). Using MRI to study monozygotic co-twins, Gilbertson and colleagues (2002) found that Vietnam veterans with PTSD were characterized by smaller hippocampi than those without PTSD, but that the co-twins of veterans with PTSD (who had not served in Vietnam) had hippocampi that were just as small. These findings suggest that small hippocampal volume may constitute a vulnerability factor for PTSD among people exposed to trauma, supported by a recent treatment study demonstrating that smaller hippocampal volume is not the result of trauma exposure and does not increase with successful PTSD treatment (van Rooij et al., 2015). There is evidence that genetic differences may influence the development of PTSD, although clear understanding of genetic risk has been hampered by small sample sizes and varying results in candidate gene studies (see Sheerin, Lind, Bountress, Nugent, & Amstadter, 2017 for a review). For example, while it is the most widely studied polymorphisms, meta-analyses of the serotonin transporter (*5-HTTLPR*) have found no overall association with PTSD, although it may be important in consideration of potential gene × environment interactions with level of trauma exposure (Armbruster et al., 2009; Gressier et al., 2013; Navarro-Mateu, Escámez, Koenen, Alonso, & Sánchez-Meca, 2013). Candidate genes involved in the regulation of the HPA axis such as *FK-506* binding protein (*FKBP5*; Watkins et al., 2016) and *PACAP* (pituitary adenylate cyclase-activating polypeptide, coded by *ADCYAP1* gene, associated with estrogen response) (Stevens et al., 2014) may be important to risk for PTSD. Genomewide association studies (GWAS) offer an unbiased approach to test associations of genetic variants across the entire genome with a construct of interest, and there have been numerous GWAS studies of PTSD to date (see Daskalakis, Rijal, King, Huckins, & Ressler, 2018, for a review). Epigenetic research on PTSD is also promising and may help us to better understand the ways that trauma exposure (among other environmental stressors) impact outcomes of DNA through changes to the epigenome, which then confer risk for the development of PTSD (Howie, Rijal, & Ressler, 2019).

It is increasingly clear that function of the inflammatory system is also important to consider in relation to PTSD (Michopoulos, Powers, Gillespie, Ressler, & Jovanovic, 2017). In the context of trauma exposure, the immune system is activated and with chronic exposure can become dysregulated; immune function and HPA axis function are closely interconnected, and the dysregulation of both are implicated in PTSD. While accumulative evidence shows immune dysregulation in PTSD (Passos et al., 2015), the directionality of the association remains unclear. Numerous prospective studies have shown that higher levels of inflammation pretrauma are associated with increased risk of PTSD posttrauma (see Michopoulos et al., 2017, for a review).

There is also evidence that people's capacity for conditioning may be a risk factor for PTSD development. One study compared startle responses in pairs of Vietnam combat veterans and their non-combat-exposed monozygotic twins (Orr et al., 2003). Orr and colleagues (2003) found evidence of more slowly habituating skin conductance to startle responses in veterans with PTSD and their non-combat-exposed co-twins compared to veterans without PTSD and their non-combat-exposed co-twins. This finding suggests that more slowly habituating skin conductance responses to startle stimuli may represent a pretrauma vulnerability factor for PTSD. A prospective study of newly recruited firefighters indicated that skin conductance and eyeblink electromyographic (EMG) startle responses before trauma exposure in a cohort of trainees predicted acute stress reactions after subsequent trauma exposure (Guthrie & Bryant, 2005). Other research indicates that impaired extinction learning prior to trauma exposure is predictive of subsequent posttraumatic stress in firefighters and paramedics (Guthrie & Bryant, 2006).

TREATMENT

Chronic PTSD

The treatment of choice for chronic PTSD is trauma-focused psychotherapy. Comprehensive treatment based on systematic reviews of the evidence including guidelines from the U.K. National Institute for Clinical Excellence (2018), the U.S. Veterans Department/Department of Defense (VA/DoD, 2017), the U.S. Institute of Medicine (2011), the Australian National Health and Medical Research Council (Phoenix Australia, 2013), the American Psychological Association (2017), and the International Society for Traumatic Stress Studies (Bisson et al., 2019) have reached this conclusion. A similar conclusion has been drawn from major meta-analyses (Bradley, Greene, Russ, Dutra, & Westen, 2005; Cusack et al., 2016). Importantly, trauma-focused psychotherapy has been shown to be efficacious across different trauma-exposed populations, including survivors of traumatic injury and assault (Bryant, Moulds, et al., 2008), sexual assault (Foa, Rothbaum, Riggs, & Murdock, 1991), combat (Castillo et al., 2016; Schnurr et al., 2003), terrorist attacks (Duffy, Gillespie, & Clark, 2007), and child sexual abuse (McDonagh et al., 2005).

The major form of trauma-focused psychotherapy, CBT, usually commences with psychoeducation about the trauma responses, then focuses on three major strategies: anxiety management, exposure, and cognitive restructuring. In psychoeducation, the patient is informed about common symptoms following a traumatic event and the way in which the core symptoms will be treated during the course of

therapy. Anxiety management techniques aim to reduce anxiety through a range of techniques that may include breathing retraining, relaxation skills, and self-talk. Exposure therapy usually involves both imaginal and *in vivo* exposure. Imaginal exposure typically requires patients to imagine their traumatic experience vividly for prolonged periods, usually for at least 30 minutes. The therapist asks patients to provide a narrative of their traumatic experience in a way that emphasizes all relevant details, including sensory cues and affective responses. *In vivo* exposure involves graded exposure to feared stimuli, in which patients are asked to remain in close proximity to mildly fearful reminders of the trauma, then repeat this exercise with increasingly fearful situations until they feel comfortable with most reminders of their experience. Exposure may be therapeutic because of habituation of anxiety, integration of corrective information, knowledge that the trauma is a discrete event that is no longer threatening, and self-mastery through management of exposure (Rothbaum & Schwartz, 2002). Cognitive restructuring involves teaching patients to identify and evaluate the evidence for negative automatic thoughts, as well as helping them to evaluate their beliefs about the trauma, the self, the world, and the future. In most controlled trials, CBT programs usually require eight to 14 sessions conducted on an individual basis.

The majority of studies indicate that treatments incorporating exposure either to trauma memories or situations that remind the individual of trauma are beneficial. There is some debate over the relative contributions of different CBT components, although evidence of the specific comparable contributions of various components of CBT remains insufficient (Cusack et al., 2016). Several studies suggest that exposure and cognitive restructuring provide comparable treatment gains, and others indicate that combining cognitive restructuring with exposure does not provide additive effects above beyond exposure alone (Foa et al., 2005). There is some evidence that the addition of cognitive restructuring may enhance the effect of exposure (Bryant et al., 2003, 2008). The overall finding that other CBT components do not markedly enhance exposure suggests that exposure is a critical ingredient in facilitating recovery from PTSD (Cahill, Rothbaum, Resick, & Follette, 2009). Because exposure typically includes a form of cognitive reframing in which the therapist discusses the mastery of exposure to trauma reminders, it seems that the change mechanisms involved in both exposure and cognitive restructuring may overlap. In summary, the largest body of evidence supports the use of exposure-based CBT that typically employs exposure to trauma memories and feared situations that result in a sense of safety or mastery over those stimuli.

Prolonged exposure (PE) is a variant of CBT and has received strong support for its effectiveness in treating PTSD across varied populations for a range of trauma types (Cusack et al., 2016; Powers, Halpern, Ferenschak, Gillihan, & Foa, 2010). PE is a manualized treatment that includes many of the components already described earlier but with one major difference, in that there is not an explicit cognitive restructuring component. The goal of PE is to alter fear structures so that they are no longer problematic; thus, the imaginal and *in vivo* exposures are focused on activating the fear structure and incorporating new corrective information into the structure (Watkins, Sprang, & Rothbaum, 2018). While PE generally is provided in eight to 15 sessions, reduced formats (five sessions in primary care settings) (Rauch, Cigrang, Austern, Evans, & Strong Star Consortium, 2017) and application in intensive outpatient

program settings (e.g., Zalta et al., 2018) have been developed and showed promise for improving outcomes.

Another variant of trauma-focused CBT, cognitive processing therapy (CPT), was developed as a form of cognitive therapy to address the specific cognitive responses of survivors of sexual assault (Resick & Schnicke, 1993); since its development CPT has shown effectiveness for a wide range of trauma types (Watkins et al., 2018). In CPT, attention is focused on reframing unrealistic beliefs about safety, trust, control, esteem, and intimacy. Originally, CPT also engaged trauma memories by requiring the patient to write detailed accounts of the trauma and relating these accounts to the therapist. However, a trial that compared (1) full CPT, (2) the writing component, and (3) the cognitive component found that the writing component did not provide any additive gain over the cognitive element (Resick et al., 2008) led to the removal of the written narrative component in the most recent manual (Resick, Monson, & Chard, 2016). CPT and PE have shown similar effectiveness, and both are recommended as frontline psychotherapy approaches for PTSD.

A third variant of CBT is eye movement desensitization and reprocessing (EMDR). This treatment differs from CBT insofar as it requires patients to focus their attention on a traumatic memory, while simultaneously visually tracking the therapist's finger as it is moved across their visual field, then to engage in restructuring of the memory (Shapiro, 1995). The patient is asked to identify more adaptive or positive thoughts related to the memory or traumatic experience, and to again track the therapist's fingers. Although the specific mechanisms by which eye movements are purported to enhance treatment gains are not well specified, moving one's eyes after engaging the trauma memory is intended to facilitate information processing (Shapiro & Maxfield, 2002). Across a series of controlled trials, EMDR has been shown to be an effective treatment for PTSD relative to wait list and to perform comparably to standard CBT. Accordingly, it is recommended, along with trauma-focused CBT, PE, and CPT, as a frontline treatment in a number of treatment guidelines (Phoenix Australia, 2013; VA/DoD, 2017). In evaluating EMDR, however, it is important to note that eye movements are not pivotal in the treatment effects of EMDR (Phoenix Australia, 2013). This conclusion is supported by evidence from studies that manipulated eye movement, in addition to other strategies, and found that the eye movements themselves were not important in relieving the symptoms relative to the other components of EMDR (Foley & Spates, 1995). Additionally, the Australian National Health and Medical Research council (NHMRC) guidelines concluded that EMDR was as efficacious as CBT only when EMDR also included *in vivo* exposure (Phoenix Australia, 2013).

There has been concern that CBT techniques may not be appropriate for patients with symptoms consistent with ICD-11 CPTSD. On the premise that patients with severe emotion regulation deficits may have difficulty managing the emotional demands of exposure-based therapy, attempts have been made to prepare these patients for trauma-focused CBT approaches by teaching them skills to regulate emotions and improve interpersonal relations. These skills involve distress tolerance, labeling emotions, and emotion management, as well as interpersonal components related to maladaptive schemas, assertiveness, and flexibility in interpersonal engagement. One approach designed specifically to treat CPTSD is skills training in

affective and interpersonal regulation (STAIR; Cloitre, Koenen, Cohen, & Han, 2002). Two randomized trials have now demonstrated that this modified form of CBT is efficacious in treating CPTSD, and there is evidence that it is more effective than standard CBT for patients with CPTSD (Cloitre et al., 2010). Considering that people who suffer more severe traumatic exposures, and often endure prolonged childhood trauma, can have difficulty regulating emotions, this approach appears to augment the efficacy of CBT. Overlap exists between CPTSD and borderline personality disorder, and dialectical behavior therapy with PE (DBT-PE) has shown promise in improving outcomes for individuals with comorbid borderline personality disorder and PTSD (Harned, Korslund, Foa, & Linehan, 2012). It is important to acknowledge that while research regarding STAIR continues (Oprel et al., 2018), current evidence for a phase-based approach to treating CPTSD is limited and general guidelines remain for the use of frontline treatment approaches (De Jongh et al., 2016), while balancing patient preferences and service availability.

An important caveat exists for the support for trauma-focused CBT. Although it is without doubt the best treatment we have for PTSD, and most patients respond positively, between one-third and one-half of patients are not responsive to this approach and continue to experience ongoing symptoms (Bradley et al., 2005; Loerinc et al., 2015). This has led to many attempts to augment trauma-focused CBT, and these have typically tried to promote extinction processes in combination with exposure therapy. Pharmacological agents to achieve this have included the partial glutamatergic agonist D-cycloserine (DCS), which has yielded mixed findings in enhancing the benefits of trauma-focused CBT (de Kleine, Hendriks, Kusters, Broekman, & van Minnen, 2012; Litz et al., 2012; Rothbaum et al., 2014); glucocorticoid administration to facilitate extinction, which has been shown in one pilot study to augment trauma-focused CBT (Yehuda et al., 2015) but has not been proven in full trial; yohimbine, which was found in one study that combined administration with trauma-focused CBT to yield no effect for yohimbine at follow-up; and MDMA (3,4-methylenedioxymethamphetamine), which has been shown in several pilot studies to augment the effects of trauma-focused CBT but not replicated in proper trials (Mithoefer et al., 2018; Ot'alora et al., 2018). Other studies have employed brain stimulation techniques, such as transcranial direct current stimulation and transcranial magnetic stimulation, but they currently lack adequate evidence (Kozel et al., 2018). Overall, although some of these approaches appear to show promise and deserve further investigation, the current evidence does not support these approaches (Lebois, Seligowski, Wolff, Hill, & Ressler, 2019).

CBT for Children

Compared to studies of adults, relatively less attention has been given to treatment of childhood PTSD. In general terms, children appear to experience PTSD as often as adults. Some differences across developmental stages have been shown, particularly in young children (Scheeringa, Myers, Putnam, & Zeanah, 2012). In DSM-5 (American Psychiatric Association, 2013), PTSD diagnosis is the same for children over the age of 6, although some of the symptoms allow for different presentation (e.g., repetitive play with themes or aspects of the trauma as a component of intrusive

symptoms). For children age 6 and younger, there is a new developmental subtype of PTSD: PTSD in children 6 years and younger. Treatment programs for children often modify CBT by integrating strategies that teach regulation of affect or behavior, including parents in treatment, adapting treatment delivery to the developmental stage of the child, and placing strong emphasis on rapport factors between therapist and patient (Cohen, Mannarino, Deblinger, & Berliner, 2009). Consistent with the adult literature, available trials support the conclusion that trauma-focused psychotherapy, and in particular, trauma-focused CBT for children and adolescents is the treatment of choice for childhood PTSD (de Arellano et al., 2014; Dorsey et al., 2017; Jensen, Holt, & Ormhaug, 2017).

Global Early Interventions

A major debate that has persisted over recent years concerns the merits of intervening shortly after a traumatic event potentially to prevent PTSD. For many years, psychological debriefing has been the most common approach immediately after trauma exposure, and critical incident stress debriefing (CISD) was arguably the most common variant of this approach (Mitchell, 1983). In the original iteration of this strategy, a debriefing session that typically occurred within 48 hours of the trauma involved education about trauma reactions; required participants to describe what occurred and their cognitive, emotional, and physical responses to the event; and provided suggestions for stress reduction (Mitchell & Everly, 1996). Although difficult to assess because of the many uncontrollable factors that exist in a postdisaster setting, a series of studies have compared trauma survivors who received CISD and those who did not receive the intervention (Rose, Brewin, Andrews, & Kirk, 1999). Systematic reviews of these studies indicate that people who receive CISD did not have better outcomes than those who do not (Rose, Bisson, & Wessely, 2001). Furthermore, some tentative evidence indicates that very distressed people who receive debriefing immediately after a trauma may subsequently have worse PTSD symptoms than those who do not receive debriefing (Hobbs, Mayou, Harrison, & Worlock, 1996). It has been suggested that requiring people to ventilate their emotions within days of trauma exposure may hasten arousal and strengthen their trauma memories, which may impede natural recovery (McNally, Bryant, & Ehlers, 2003). In the wake of this evidence, the field has moved toward recognition that single-session interventions after trauma may not be sufficient to avert subsequent PTSD. At present, the most common alternative to debriefing is psychological first aid (PFA; Young, 2006), which attempts to assist adaptation by ensuring that patients' needs are met, facilitating problem solving, reducing acute arousal, and accessing social supports. PFA attempts to enhance coping skills in the immediate aftermath of a disaster *without* causing undue distress by directing the survivor to revisit the traumatic experience. Importantly, PFA has not yet been tested, and one obstacle to evaluation is that PFA does not have clearly stated goals or outcomes. Therefore, it is not a standardized intervention that is amenable to controlled evaluation (Dieltjens, Moonens, Van Praet, De Buck, & Vandekerckhove, 2014; Fox et al., 2012). Whereas CISD explicitly attempts to reduce subsequent PTSD, PFA assists coping rather than serving a secondary prevention role.

Targeted Early Interventions

In contrast to interventions provided to all trauma survivors shortly after exposure, there has also been a focus on interventions for survivors who are deemed to be at high risk for developing PTSD. The focus of most of these studies has been on trauma survivors who display ASD in the first month after trauma exposure because of evidence that these individuals are more likely than other survivors to develop chronic PTSD (Bryant, 2003). Attempts at secondary prevention have typically modified standard CBT approaches by abridging them to five or six sessions. The common factor across interventions for ASD/acute PTSD is that their trauma-focused exposure approach involves imaginal or *in vivo* exposure. Across a series of controlled trials, there is now convergent evidence that exposure-based CBT provided within a month of trauma exposure prevents PTSD in the majority of cases relative to nondirective counseling (Bryant, Mastrodomenico, et al., 2008). These treatment gains have been shown to maintain up to four years after treatment (Bryant et al., 2006). Additionally, providing CBT to trauma survivors with acute PTSD in the initial months after trauma exposure results in reduced levels of PTSD at subsequent time points (Ehlers et al., 2003). There is also preliminary evidence of the benefits of a three-session modified PE for individuals immediately posttrauma, with the intervention initiated bedside in the Emergency Department (Rothbaum et al., 2012). Since many individuals show natural recovery, it is critical that reliable early risk factors for PTSD directly following trauma exposure in Emergency Department settings are identified to improve targeted early intervention approaches (Maples-Keller et al., 2020). Overall, the evidence indicates that exposure-based therapies are indicated for early intervention with people with ASD/acute PTSD.

CLINICAL IMPLICATIONS AND PRINCIPLES OF CHANGE

This review has highlighted the heterogeneity of PTSD. The potential variability and complexity of PTSD highlights how clinicians need to be cognizant of the various techniques from evidence-supported research that can be administered to posttraumatic presentations. One of the first decisions is whether a patient is appropriate for exposure-based therapy. Although it is the treatment of choice, some patients may not be able to manage it appropriately for several reasons. Patients who are potentially psychotic, severely suicidal, or substance dependent may be better managed initially with medication, skills-based interventions (e.g., STAIR, PE), or programs specifically designed for comorbid PTSD and substance use disorders (i.e., Seeking Safety), until these conditions are adequately controlled. Any condition that may undermine extinction learning during exposure should be managed prior to exposure.

Table 6.1 presents a summary of principles and related strategies that can be applied to the variety of PTSD presentations, and the sequence in which one implements these strategies is usually determined by several factors. As always, with any clinical intervention, any presentation that poses a risk for a patient (e.g., self-harm, reckless behavior) needs to be addressed first.

TABLE 6.1. Principles of Change and Treatment Choices

Principle	Treatment strategy
Understanding of trauma reactions	Psychoeducation and normalization of symptoms
Enhancing distress tolerance	Emotion regulation skills training (breathing retraining, emotion labeling, mindfulness)
Extinction learning of trauma memory	Prolonged imaginal exposure
Extinction of situational reminders	*In vivo* exposure
Modifying maladaptive appraisals of the trauma or one's response	Cognitive restructuring
Managing harmful substance use	Initially provide exposure to reduce source of self-medication. Give medication and implement behavioral control and motivational interviewing to address substance abuse
Reducing sleep disturbance	Address nightmares through imaginal exposure and, if necessary, implement a sleep hygiene regimen to reduce sleep-incompatible behaviors
Reducing rumination on trauma or one's response	Prolonged exposure to minimize ruminative response, motivational interviewing to understand the impact of rumination, and distraction or mindfulness to manage rumination
Managing anger	Motivational interviews concerning the benefits and costs of anger, cognitive restructuring of beliefs driving anger, and mindfulness practice to reduce anger-related thoughts
Addressing comorbid depression	Prolonged exposure to reduce PTSD symptoms, cognitive restructuring to address self-esteem and hopelessness beliefs, and event scheduling to increase positive activities

Having ensured the safety of the patient, the next principle is to sequence those strategies that lead to generalized improvement. In the context of PTSD, it is often understood that many of the related problems, including depression, panic symptoms, sleep disturbance, or substance use, may be exacerbated by the symptoms of PTSD. In fact, these related problems often diminish following successful treatment of the PTSD; accordingly, treatment should focus on exposure and cognitive restructuring strategies, with the expectation that reduction of trauma-related distress will generalize to improvements in other domains. If residual problems remain, then specific techniques can be applied to these remaining problems (e.g., sleep hygiene, panic). In the case of patients who have panic disorder, the clinician who wants to reduce the likelihood of full panic attacks in the course of exposure can conduct interoceptive exposure prior to exposure.

Another common comorbidity to consider in treatment is traumatic brain injury. These cases pose a challenge for exposure because they typically involve some degree of amnesia of the event that triggered the disorder. In mild traumatic brain injury, there are usually islands of memory that allow exposure to be administered in the normal manner (Bryant, Moulds, Guthrie, & Nixon, 2003). In more severe cases of brain injury, when there is often no discursive memory of the trauma, patients often experience psychological or physiological distress in response to reminders of the trauma, without actually recalling the event (Bryant, Marosszeky, Crooks, & Gurka, 2000). In these cases, it is more useful to focus on *in vivo* rather than imaginal exposure because situational factors tend to be the major triggers of fear.

It is also crucial to determine whether the patient is ready to allocate sufficient resources to therapy. It is common in the posttraumatic period, especially in the acute phase, for the trauma survivor to be challenged with a range of environmental and personal demands, including pain, surgery, legal processes, housing disturbance, or ongoing threat. PTSD is also associated with a range of functional challenges, such as employment problems, health difficulties, marital and interpersonal problems, sleep, and sexual dysfunction; thus, systematic assessment of these clinical features should be used in order to determine whether patients can focus appropriately on therapy when they are attempting to manage the demands of their environment. There is growing support for the use of comprehensive services that include both evidence-based therapies and social work services within the same program in medical settings in which high volumes of acute trauma patients are seen (e.g., Trauma Recovery Centers). In order to respond most effectively to clients' needs and to reduce possible interference to therapy, some of these issues may need to be addressed before and/or during exposure-based therapy. These issues should also be assessed at the end of successful exposure-based treatment to determine whether additional interventions are necessary to improve clients' health and functioning.

CONCLUSION

Much has been learned about human response to psychological trauma since the introduction of the PTSD diagnosis. In the wake of large-scale natural and man-made disasters, interpersonal and community violence exposure, and the continuing effects of war, research continues to shed light on the biological, cognitive, and social factors that influence the different trajectories of response to trauma. Evidence emerging from these studies continues to shape policy and practice to enhance the ways we attempt to prevent, assess, and treat PTSD. There is no doubt that traumatic experiences will continue to affect people at a very frequent rate, and it is critical that we constantly adopt scientific approaches to help people respond to these events in ways that enhance adaptation.

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CHAPTER 7

Eating Disorders

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This chapter presents an overview of the clinical features and implications, risk factors, and treatment of four eating disorders: anorexia nervosa (AN), bulimia nervosa (BN), binge-eating disorder (BED), and avoidant/restrictive food intake disorder (ARFID).

TYPICAL SYMPTOMS, COURSE, COMORBIDITY, AND ASSOCIATED FEATURES

Anorexia Nervosa

AN is characterized by failure to maintain a minimally acceptable weight (e.g., a body mass index [BMI] of less than 18.5 kg/m^2, failure to meet expected growth in youth) and intense fear of being or becoming fat. The revision of the World Health Organization's (2019) *International Statistical Classification of Diseases and Related Health Problems* (ICD-11) also notes that rapid weight loss (e.g., >20% of total body weight within 6 months) may satisfy the low weight guideline as long as other diagnostic requirements are met (Claudino et al., 2019). Patients with AN also place undue value on their own body shape or weight, and they deny that their low weight is a serious health concern. Females with AN commonly experience an interruption in menstrual activity, but as this does not meaningfully distinguish clinical status, it is not an essential diagnostic feature. There are two subtypes of AN—one for patients who regularly engage in binge-eating or purging behaviors while underweight, and the other for patients who simply restrict their food intake. ICD-11 also includes a severity qualifier to ensure that individuals can retain the AN diagnosis until a "full and lasting recovery is achieved" (e.g., "healthy weight and the cessation of behaviors

leading to weight loss independent of the provision of treatment"; Claudino et al., 2019).

Psychological symptoms associated with AN include depression, anxiety, insomnia, and increased obsessionality. These symptoms have also been observed in non-eating-disordered individuals undergoing an induced state of semistarvation and in the case of AN are often secondary to underweight status (Attia, 2010). Medical complications of AN include cardiac abnormalities, osteopenia or osteoporosis, and endocrine disturbances. Electrolyte imbalances may be present, including hypokalemia in individuals with vomiting or laxative use, and hyponatremia in those who ingest large amounts of liquids (Attia, 2010).

AN is a serious psychiatric illness associated with increased morbidity and mortality. Over the long term, while approximately one-half of the individuals with AN achieve full recovery, approximately one-third remain partially symptomatic, and approximately one-fifth remain chronically ill. A meta-analysis of 35 published studies describing mortality rates for AN determined the crude mortality rate to be 5.1 deaths per 1,000 person-years (Arcelus, Mitchell, Wales, & Nielsen, 2011). Mortality estimates rise for patients who require hospitalization, with as much as a sixfold increase in comparison to the general population. The cause of death in AN is equally or more likely to be suicide than to be medical problems secondary to compromised nutritional status (Papadopoulos, Ekbom, Brandt, & Ekselius, 2009).

High rates of depression, dysthymia, social phobia, generalized anxiety disorder, specific phobia, and obsessive–compulsive disorder (OCD) are commonly reported in clinical samples (American Psychiatric Association, 2013). Individuals with AN report elevated lifetime alcohol use disorder that typically begins after onset of AN; however, the rate of alcohol use disorder for individuals with AN, restricting subtype (AN-R), is significantly lower than for individuals with AN, binge-eating/purging subtype (AN-B/P), BN, or BED (Bulik et al., 2004).

AN is associated with personality traits including perfectionism, obsessive–compulsiveness, neuroticism, negative emotionality, harm avoidance, low self-directedness, low cooperativeness, and avoidant personality disorder (Cassin & von Ranson, 2005). Individuals with AN-R report high levels of constraint and persistence and low levels of novelty seeking, though these features are not ubiquitous. Perfectionism and harm avoidance are traits that appear to be premorbid to AN, enduring even after recovery (Strober et al., 2006).

AN can also be associated with a range of psychosocial problems. Significant interpersonal problems or stressors often occur in the year prior to onset of AN and cumulatively increase risk for AN (Pike et al., 2008). Individuals with AN often report social isolation and lack of pleasure in interpersonal relationships, and the majority of women with AN report dramatically decreased sexual interest and pleasure (Zemishlany & Weizman, 2008). Adults with AN are likely to report lower rates of employment than non-eating-disordered peers given the all-consuming nature of the disorder. This is particularly true during the acute stages of the disorder; however, for individuals with long-standing AN and for those with significant comorbid psychopathology, employment remains a problem even following recovery (Hjern, Lindberg, & Lindblad, 2006).

For many, the impact of the disorder can be pervasive and long-lasting; however, most follow-up studies of AN have been conducted with individuals who pursued

intensive treatment and are therefore likely to represent a more severely ill subset. Among those with AN who develop their eating disorder in adolescence, and for whom the disorder is not enduring, associated psychopathology and related problems are accordingly more limited. Even for those with more long-standing AN, certain interpersonal problems and personality dimensions (and even disorders) can improve with recovery from AN. Thus, clinicians should be careful not to overinterpret personality disorders and interpersonal problems during the acute stage of AN, but rather should monitor an individual's clinical presentation, including these features, and address them as an integrated part of the clinical work.

Bulimia Nervosa

Individuals with BN engage in repeated binge-eating episodes and use compensatory behaviors, such as self-induced vomiting or taking laxatives or other pills, over at least a period of several months. An episode of binge eating involves taking in an objectively large amount of food (i.e., an amount that most people would agree is large) *and* simultaneously experiencing a sense of loss of control over eating. According to ICD-11, it is not necessary to consume an objectively large amount of food during a binge-eating episode; this definitional change is likely to expand the number of individuals diagnosed with BN (and reduce the number with an other specified feeding and eating disorder; Claudino et al., 2019). Patients with BN overvalue body shape and weight. The medical complications associated with BN are primarily related to the deleterious effects of purging. The most common problems are erosion of dental enamel, swollen salivary glands, edema, electrolyte abnormalities, and esophageal tears (Mehler, Birmingham, Crow, & Jahraus, 2010).

Given the lack of agreement on definitions of remission, recovery, and relapse in BN (Shapiro et al., 2007), the evaluation of outcome for this patient population remains problematic. A longitudinal study reported that approximately 75% of female college students diagnosed with BN by structured interview were in remission at 20-year follow-up (Keel & Klump, 2003). In treatment-seeking samples, between 40 and 45% of patients achieve full recovery, 27% experience symptom improvement, and 14–23% remain chronically ill in the long term (Quadflieg & Fichter, 2019; Steinhausen & Weber, 2009). The most common diagnostic crossover is from BN to another specified (or unspecified) feeding and eating disorder, followed by AN, then BED (Quadflieg & Fichter, 2019; Steinhausen & Weber, 2009). Crude mortality rates for individuals with BN are generally lower than those in AN (Papadopoulos et al., 2009), with one meta-analysis finding the crude mortality rate in BN to be 1.7 per 1,000 person-years (Arcelus et al., 2011).

Among individuals with a lifetime history of BN, mood and anxiety disorders are common; approximately 43–75% of patients with BN report comorbid mood disorders (Duncan et al., 2005; Fichter & Quadflieg, 2004), while approximately 68% experience some form of anxiety or OCD (Kaye, Bulik, Thornton, Barbarich, & Masters, 2004). Depressive symptoms endorsed by individuals with BN may include distress about binge-eating episodes, and negative affect or emotion dysregulation may also be associated with the onset of aberrant eating (Fischer, Smith, & Cyders, 2008; Roberto, Grilo, Masheb, & White, 2010). Compared to people with AN, those with BN are more likely to report a lifetime history of an alcohol use disorder (46.1%); similar

to AN, the majority of individuals with BN who develop an alcohol use disorder do so subsequent to the onset of their eating disorder (Bulik et al., 2004).

A high percentage of individuals with BN also meet criteria for comorbid personality disorders, with borderline and avoidant personality disorders being most common (Rø, Martinsen, Hoffart, & Rosenvinge, 2005). Impulsivity and, more specifically, negative urgency, is commonly associated with BN and is significant in predicting treatment outcome (Fischer et al., 2008). Suicide attempts are not uncommon in individuals with BN, with one study reporting that 26.9% of women with a lifetime history of BN have attempted suicide. However, the rate of completed suicides for individuals with BN is reported to be lower than that for those with AN.

Individuals with BN report significant difficulties in relationships, and interpersonal problems are correlated with treatment outcome (Hartmann, Zeeck, & Barrett, 2010). However, in comparison to those with AN, individuals with BN are much more likely to be able to continue working and to be socially engaged in their peer communities, even while symptomatic.

Binge-Eating Disorder

BED is characterized by recurrent episodes of binge eating that occur in the absence of any compensatory behaviors. As in BN, there is a disagreement about diagnostic standards, with DSM-5 requiring the presence of objectively large binge episodes for a BED diagnosis, but ICD-11 including intake of both objective and subjective amounts of food with loss of control (Claudino et al., 2019). Binge eating in the context of BED is accompanied by significant emotional distress and is further distinguished from normal eating by features such as eating quickly, until uncomfortably full, or when not hungry. BED has no diagnostic criterion specifying the cognitive feature of undue influence of body shape or weight on self-evaluation. However, individuals with BED tend to report more concern with body shape and weight than overweight (Grilo, Masheb, & White, 2010) and normal-weight (Goldschmidt et al., 2010) controls. The degree of overvaluation reported by BED patients is similar to that reported by individuals with BN (Grilo et al., 2010). Body shape and weight concerns indicate a more severe variant of BED associated with greater psychosocial impairment (Goldschmidt et al., 2010).

Individuals with BED are more likely than those without the disorder to be obese (Hudson, Hiripi, Pope, & Kessler, 2007), but they can also be distinguished reliably from non-eating-disordered obese individuals (Wonderlich, Gordon, Mitchell, Crosby, & Engel, 2009). When compared to weight-matched individuals without BED, those with BED endorse significantly more eating disorder psychopathology and lower quality of life (Wonderlich et al., 2009). BED is associated with co-occurring medical problems such as diabetes, irritable bowel syndrome, and fibromyalgia (Olguin et al., 2017), and psychological correlates, such as depressive symptoms and low self-esteem (Hrabosky, Masheb, White, & Grilo, 2007). A meta-analysis of BED studies from 2004 to 2009 reported a cumulative mortality rate of 0.5%, but more data are needed from studies with longer follow-up periods (Keel & Brown, 2010).

The course of BED appears quite distinct from that of AN, and although BED shares core features with BN and diagnostic crossover can occur, there are also clear differences in risk, course, and outcome. Individuals with BED who receive

treatment are more likely to experience symptom reduction and less likely to experience diagnostic crossover (Wonderlich et al., 2009).

Compared to obese controls, individuals with BED show higher rates of mood and personality disorders, with depression as the most commonly reported (lifetime rate of 50–60%) (Wonderlich et al., 2009). Negative affect, state anxiety, anxiety disorders, and alcohol use disorders are also elevated in clinical samples (Dingemans, Danner, & Parks, 2017). The degree of psychopathology reported by individuals with BED appears to be associated with the eating disorder, not degree of obesity (Wilfley, Citrome, & Herman, 2016).

Avoidant/Restrictive Food Intake Disorder

ARFID, as its name suggests, is an eating disorder defined by highly restrictive eating and is associated with significant medical and psychosocial consequences. Added to the eating disorders section in DSM-5 (American Psychiatric Association, 2013) and soon to be added to ICD-11 (Claudino et al., 2019), ARFID is a reformulation of DSM-IV's and ICD-10's feeding disorder of infancy and early childhood (American Psychiatric Association, 1994). The age-of-onset criterion was removed because individuals of all ages can have clinically significant avoidant/restrictive eating, and it does not always begin in childhood. Unlike those with AN, individuals with ARFID attribute their food avoidance or restrictive eating to sensory features (e.g., texture, color), fears related to the process of eating (e.g., choking, vomiting), and/or a lack of interest (e.g., low hunger, lack of pleasure of eating) rather than body shape or weight concern. ARFID is not synonymous with picky eating. Though roughly 25% of youth experience feeding or eating problems (Benjasuwantep, Chaithirayanon, & Eiamudomkan, 2013), ARFID is distinguished by its severity and persistence.

Although DSM-5 offers a clear definition of ARFID, this diagnostic group is rather heterogeneous. ARFID patients appear more likely to be younger, male, and to have a higher incidence of comorbid psychiatric and medical disorders compared to those with other eating disorders (Norris, Spettigue, & Katzman, 2016). Common comorbid psychiatric disorders shared with AN, BN, and BED include anxiety disorders, OCD, and depression. Unlike the other eating disorders, autism spectrum disorders, oppositional defiant disorder, and attention-deficit/hyperactivity disorder may more commonly co-occur.

EPIDEMIOLOGY

Global Burden of Disease and Prevalence

It is widely recognized that eating disorders occur across culture, socioeconomic class, and race (Claudino et al., 2019), with risk being a function of biological, environmental, cultural, and psychological factors. The Global Burden of Disease (GBD) study has dramatically enhanced our understanding of the worldwide distribution and burden of eating disorders. Utilizing the metric of disability-adjusted life-years (DALYs), the GBD study provides rates of hundreds of health conditions and an estimate of burden based on an algorithm that incorporates years lived with the disorder, years of life lost due to the disorder, and associated degree of impairment.

Current GBD estimates comparing AN and BN to 306 other health conditions indicate that combined, these disorders were the 12th leading cause of DALYs in females ages 15–19 in high-income countries, responsible for 2.2% of all DALYs. Globally, this ranking did not change a great deal between 1990 and 2013; however, within low- and middle-income countries, AN and BN increased from 58th to 46th (Erskine, Whiteford, & Pike, 2016). This higher ranking is largely a function of improvements in preventing and treating other communicable diseases in these countries, which shifted the proportion of total burden attributed to eating disorders (Hoek, 2016). The number of DALYs per 100,000 members of the population is highest in Europe (Erskine et al., 2016). Despite lower prevalence rates, given the large populations of India and China, along with the United States, these countries have the highest contributions of total DALYs caused by eating disorders among women ages 15–49 (Thomas, Lee, & Becker, 2016).

Epidemiological estimates are largely derived from studies in Western Europe and North America. Epidemiological data from Asia and Pacific Island countries remain scarce (Thomas et al., 2016); the same holds true for Latin America (Kolar, Rodriguez, Chams, & Hoek, 2016) and even more so for Africa (van Hoeken, Burns, & Hoek, 2016).

Anorexia Nervosa

AN, while a severe psychiatric illness, is also relatively uncommon, and rates appear to be relatively stable in high-income countries, with the exception of a noted increase among females ages 15–19 (Smink, van Hoeken, & Hoek, 2013). Estimates of incidence and prevalence of AN may be complicated by the rarity of the illness and patient characteristics (i.e., secrecy/denial around symptoms that may interfere with accurate reporting; Grilo, 2006). With changes in the formal diagnostic criteria of AN, individuals who had previously met criteria for eating disorder not otherwise specified (EDNOS) are now more likely to be diagnosed with AN, but the overall rates do not actually appear to be increasing (Mustelin et al., 2016). The most recent studies in Europe show that AN affects 1–4% of women in Europe; in the United States, the prevalence rate is approximately 0.8% in adults and 0.3% in adolescents (Swanson, Crow, Le Grange, Swendsen, & Merikangas, 2011; Udo & Grilo, 2018). Best estimates indicate the disorder affects one-tenth as many males as females (Udo & Grilo, 2018). Rates of AN remain low in Africa (van Hoeken et al., 2016), consistent with the absence of cases in both a large-scale prevalence study among Black women in the United States (Striegel-Moore et al., 2003) and a large study of incidence among the majority Black population of Curaçao (Hoek et al., 2005; van Hoeken et al., 2016).

Bulimia Nervosa

Estimates of the prevalence of BN suggest it is a slightly more common disorder than AN. Hoek and van Hoeken's (2003) epidemiological review described a 1.0% aggregated prevalence rate for BN (range = 0.0–4.5%), with higher rates for more broadly defined BN (Hail & Le Grange, 2018). Studies based on surveys of large, nationally representative, U.S. samples have yielded similar rates in adolescents and lower rates (0.28%) in adults, with a slightly higher ratio of men affected than seen in AN

(Swanson et al., 2011; Udo & Grilo, 2018). The wide variation of prevalence estimates may be attributed to changing definitions of the illness since its introduction (Crow & Brandenburg, 2010). Given the secrecy and ease of hiding bulimic behaviors, incidence rates are generally considered to be underestimates (Grilo, 2006); however, there does appear to be reduced incidence over the past few decades (Smink et al., 2013).

Nationally, research suggests that BN may be more prevalent among Latinos and Blacks than non-Latino Whites (Marques et al., 2011). Associated distress with bulimic symptoms may vary cross-culturally, with preliminary data suggesting binge eating as the best predictor of distress among Whites, Blacks, and Latinos, in contrast with vomiting as the most salient predictor of distress among Asians (Franko, Becker, Thomas, & Herzog, 2007). Cross-cultural variations support the development of flexibility within the classification system to promote cultural sensitivity and allow for phenomenological distinctions (Becker, 2007).

BN has also been described in a variety of non-Western cultures (Nobakht & Dezhkam, 2000; Pike & Mizushima, 2005). Unlike in AN, the transmission of Western ideals (e.g., thin beauty) and economic factors (e.g., an abundance of food available for binge eating) is thought to play an important role in the prevalence of BN in non-Western cultures (Keel & Klump, 2003).

Binge-Eating Disorder

BED is estimated to affect 1.6% of adolescents and between 0.8 and 2.8% of adults (Hudson et al., 2007; Swanson et al., 2011; Udo & Grilo, 2018). In addition to being more common than either AN or BN, BED is associated with later age of onset and longer duration (Hudson et al., 2007). Full and subthreshold BED is more commonly seen in adolescent (0.8–2.6%) and adult (1.9–2.0%) males than is AN or BN, though the available data indicate that full-syndrome BED is still more common in women (Udo & Grilo, 2018). Prevalence estimates range from 2 to 5% in obese community samples, and as high as 30% among individuals seeking bariatric surgery or alternative weight loss treatments (de Zwaan, 2001; Kalarchian et al., 2007). In a sample of overweight and obese veterans seeking assistance with weight management, estimates were higher, with 78.2% of individuals reporting binge eating, though these findings are based on self-report data and likely capture both sub- and full-threshold symptoms (Higgins et al., 2013). Relative to other eating disorders, BED is known to be more prevalent across a variety of racial and ethnic groups (Wonderlich et al., 2009).

Avoidant/Restrictive Food Intake Disorder

There has been little epidemiological research on ARFID to date. In a population-based survey of individuals at least 15 years old, the 3-month point prevalence of ARFID was 0.3% (Hay et al., 2017), placing it in a range comparable to AN and BN. In a school-based survey of children ages 8–13 in Switzerland, 3.2% of students reported features consistent with a diagnosis of ARFID (Kurz, van Dyck, Dremmel, Munsch, & Hilbert, 2015). Medical record review studies have looked at the rates of ARFID in specialty services, reporting estimates ranging from 5 to 22.5% in pediatric or adolescent medicine eating disorders programs (Nicely, Lane-Loney,

Masciulli, Hollenbeak, & Ornstein, 2014; Norris et al., 2014), and a 1.5% prevalence rate in a pediatric gastroenterology specialty clinic (Eddy et al., 2015). It appears that a greater number of males than females are affected by ARFID relative to AN and BN. Even less is known about the prevalence of ARFID across different racial and ethnic groups. However, the disorder has been described in adolescents and adults receiving eating disorders treatment in Japan (Nakai, Nin, Noma, Teramukai, & Wonderlich, 2016; Tanaka et al., 2015).

ETIOLOGY

The empirical literature on the etiology of eating disorders is still limited, with few reports utilizing prospective and experimental research designs that could identify directional and causational effects. Several case–control studies have identified risk correlates (Fairburn et al., 1998; Karwautz et al., 2001; Pike et al., 2008; Striegel-Moore et al., 2005). These studies offer the advantage of examining actual clinical cases, but they are subject to potential reporting bias given that the data are gathered retrospectively from individuals with eating disorders. However, the majority of prospective and experimental studies of eating disturbance are limited because they do not examine true cases of eating disorders (for a review, see Jacobi, Hayward, de Zwaan, Kraemer, & Agras, 2004).

In 1986, Striegel-Moore, Sliberstein, and Rodin posed the following set of heuristic questions to guide the study of the development of eating disorders: Why now? Why women? Why some women and not others? These questions reflect several important understandings of risk for eating disorders (Striegel-Moore et al., 1986). First, eating disorders are a contemporary problem; prevalence rates have increased during the past century in Western cultures, and rates of eating psychopathology continue to increase across the globe (Becker, Thomas, & Pike, 2009). Second, gender is perhaps the most significant and unequivocal risk factor for eating disorders, particularly AN and BN. Finally, despite widespread sociocultural and environmental factors, only a small percentage of individuals actually develop significant eating disturbances, implying that individual risk factors are essential to understanding why some individuals develop eating psychopathology while others from the same environmental context do not.

Why Now?

At the broadest level, sociocultural factors set the general stage of risk. Sociocultural studies of the development of eating disorders suggest a complex interplay between increased abundance in the food supply (especially highly palatable, caloric foods), increased sedentary lifestyles, and societal overvaluation of a thin beauty ideal (Brownell, 2010). In addition, as societies modernize and industrialize, traditional gender roles are often put in flux. It has been argued that the dynamic of social transition increases risk for eating disorders at the broadest level because such societal flux increases the developmental challenges of coming of age given the lack of role models and societal norms to guide the course (Katzman & Lee, 1997).

Why Women?

In addition to biological vulnerabilities that likely predispose women to develop eating disorders, environmental factors associated with industrialization and modernization have profoundly impacted women's place in society. Across the globe, issues of women's roles and rights have been at the forefront of much societal change, and although this has ushered forth many positive changes in women's roles, a concomitant phenomenon has been the rise in societal overvaluation of a thin beauty ideal for women. The confluence of these factors results in significant pressure conveyed by family, peers, and mass media to achieve an unrealistic ideals within a social context associated with increased caloric consumption and reduced energy expenditure. To the extent that an individual internalizes the thin beauty ideal, the tension that ensues has been shown to be associated with increased body dissatisfaction and dieting, and prospective studies suggest that body dissatisfaction and dieting are significant risk factors for eating psychopathology, particularly bulimic symptomatology (Stice & Agras, 1998).

Why Some Women and Not Others?

Exposure to particular life events, individual characteristics, and genetic differences and vulnerabilities have been examined in an attempt to identify why particular individuals are at increased risk for the development of eating disorders.

The genetics data suggest some biological heritability of risk for AN, BN, and BED; however, it is unclear what the genetic vulnerability is, whether it is the same across eating disorders, and what degree of variance is accounted for by genetic factors in an etiological model (Bulik, Kleiman, & Yilmaz, 2016). It appears that one biological component of risk centers on weight regulation. Individuals with AN may have a biology that can sustain extreme weight loss that the majority of the population could not tolerate. Those with BED and BN are more likely to have a history of being overweight. There are likely other biological factors that contribute to risk; however, without the existence of specific biological or genetic markers, studies must identify variables that may further be explored in terms of both biological and environmental transmission.

In terms of the development of AN, no prospective and experimental studies exist to inform the discussion of risk factors. Case–control studies suggest that perfectionism, negative affectivity, family discord and high parental demands, and childhood physical and sexual abuse are potential risk correlates in the developmental history reported by individuals in treatment for AN (Fairburn, Cooper, Doll, & Welch, 1999; Karwautz et al., 2001; Pike et al., 2008). Interestingly, individuals with AN do not report a long history of weight and shape concerns; instead, the salience of these factors appears to be greatest in the year immediately preceding the onset of their eating psychopathology. In fact, it may be that dieting and weight concern are more accurately understood as prodromal correlates of AN rather than potential risk factors. Moreover, it is possible that the risk factors identified to date are most relevant to individuals with long-standing AN given that the existing data are derived from studies of individuals with enduring pathology.

It is commonly understood that a combination of biological vulnerabilities interact with certain environmental exposures to increase risk for AN. For example, a majority of girls engage in dieting at some point during adolescence. In the case of AN, it may be that dieting and weight loss trigger a certain biological vulnerability among individuals who also have some anxious and perfectionistic traits, such that they would be able to lose and sustain significant weight loss when others could not.

In the case of BN, extant data converge to suggest that genetic vulnerability, childhood obesity, weight concern, negative body image and dieting, sexual abuse, and adverse family factors (including parental obesity, parental mood or substance abuse disorder, and highly critical parenting), along with individual factors of social anxiety, negative affectivity, low self-esteem, impulsivity, and externalizing behaviors may contribute to increased risk (Fairburn et al., 1998; Jacobi et al., 2004; Stice, 2001).

BED shares with BN the common feature of binge eating at the core of the syndrome; thus, certain risk correlates also appear to be common across these disorders. In particular, childhood obesity, adverse family factors (including family discord, high parental demands, and parental history of mood and substance disorder), mood disorder, and negative affectivity have been identified as risk correlates in case–control studies (Fairburn et al., 1998; Striegel-Moore et al., 2005). It is likely that both biological and environmental factors contribute to increased risk for childhood obesity, which is often associated with teasing and body dissatisfaction.

Of note, many of the possible risk factors associated with eating disorders may be shared with other psychiatric disorders. For example, childhood abuse increases risk for not only eating disorders but also anxiety and mood disorders (Jonas et al., 2011). Yet unlike the presence of a certain cellular abnormality that indicates without a doubt the presence of a particular cancer, the correlates associated with eating disturbances represent increased risk but do not definitively determine the presence of an eating disorder.

MULTIVARIATE MODELS OF ONSET AND MAINTENANCE OF EATING DISORDERS

It is widely recognized that eating disorders are multidetermined and that the factors contributing to their onset are not necessarily those that contribute to maintenance of the disorder. Cognitive-behavioral therapy (CBT) interventions emphasize this point (Fairburn, Marcus, & Wilson, 1993). These models maintain that certain variables likely increase risk during a developmentally vulnerable period of time (e.g., overvaluation of shape and/or weight), and that other variables contribute to the persistence of the eating psychopathology (e.g., strict dietary rules). Specific components of the eating disorders (e.g., severe weight loss, binge eating and purging) have significant biological, behavioral, and psychological effects that serve to complicate the clinical picture and perpetuate the disturbance. CBT and other evidence-based treatments for eating disorders largely focus on identifying maintenance factors based on the tenet that addressing these issues first results in the most significant and tangible clinical improvement initially. Resolution of the acute issues brings clarity of vision in analyzing the more distal, developmental factors because by the

time the individual presents for treatment, the eating disorder has often taken on a life of its own. Cognitive-behavioral theory maintains that achieving some symptom resolution offers immediate relief and increases the patient's sense of self-efficacy. Long-standing issues can then be more readily addressed once eating disorder maintenance behaviors have sufficiently resolved.

Although CBT was initially developed for patients with BN, Fairburn introduced an enhanced version of CBT (CBT-E) proposed to be "transdiagnostic," in that it focuses on commonalities among the eating disorders (Fairburn, 2008). The core treatment is similar to that in the earlier manual (Fairburn, Marcus, et al., 1993), with two substantial changes. First, there is a reformulated strategy and method for addressing dysfunctional thought processes regarding the overvaluation body shape and weight. Second, a new treatment module tackles "mood intolerance" as a trigger of aberrant eating episodes. The core focus on eating disorder symptoms is supplemented by flexible, personalized use of modules to address maintaining mechanisms theorized to perpetuate the illness, including low self-esteem, clinical perfectionism, and interpersonal difficulties. Given the additional mechanisms addressed throughout treatment, CBT-E is proposed to be a more suitable treatment for particularly complex cases (de Jong, Schoorl, & Hoek, 2018). As we discuss in more detail later, CBT and CBT-E have strong empirical support in the field.

CLINICAL IMPLICATIONS, TREATMENT PLANNING, AND EVIDENCE-BASED TREATMENT OF EATING DISORDERS

Anorexia Nervosa

Individuals with AN are unlikely to present with complaints about their change in weight or nutritional intake. They may be identified on routine medical assessment or may present for evaluation following concern expressed by family, friends, coworkers, or coaches. Evaluation is challenging because of the thought content frequently associated with the illness, including a belief that one is fat or would become fat with weight gain, and a denial of the seriousness of the low weight state. Life events and psychological traits that were present before the onset of eating behavioral change are part of a comprehensive evaluation, but the clinician should use caution in making any causal hypotheses regarding the development of the eating disorder. Evaluation should include detailed assessment of daily food and beverage intake, weight history (e.g., lifetime highest–lowest weights; time course of significant weight changes; associated changes in menstrual status), body image concerns, and exercise routines. Assessments should include direct questioning about purging behaviors, including vomiting, as well as the use of laxatives, diuretics, and diet pills in traditional pharmaceutical, as well as alternative, preparations. Parallel history from family members may be necessary. In younger patients, weight loss may not be present; rather, the diagnosis may be made if expected goals for weight gain and growth are not met (Rosen, 2010). Evaluation will likely include medical assessment, including vital signs such as heart rate, blood pressure, and temperature, in addition to height and weight. Low weight patients will frequently have changes consistent with a hypometabolic state, and both these findings and other physical signs can be

useful indicators of the severity of illness. Some patients are concerned enough about these changes that they are willing to consider engaging in weight restoration. Even in the context of significant low weight, laboratory assessments may not identify abnormalities. Clinicians should not be reassured by these results because total body levels of micronutrients and minerals may not be reflected in blood tests.

Case formulation should include a narrative that helps the patient understand the emergence and perpetuation of symptoms. The narrative should include the patient's predisposing factors such as preillness anxiety or obsessionality, family history of an eating disorder, dieting behaviors, or activities (e.g., athletics, modeling) that may reinforce leanness. Dietary changes, even those that the patient dismisses as distinct from the formal eating disorder, such as lists of disliked foods, recently acquired vegetarian practices, gluten sensitivity and food allergies, should be considered as possible symptoms of the illness itself. Patients should be informed about the physical and psychological risks of malnutrition, including the development or intensification of distressing symptoms such as depression, anxiety, and social isolation. Basic principles of nutritional rehabilitation should be reviewed, such as the fact that individuals usually require 1,800–2,400 kcal (kilocalories) for weight maintenance, and that 3,500 kcal above maintenance requirements are necessary in order to gain 1 pound. Patients (and families) may have to hear this repeatedly in order to process the fact that nutritional plans must include 500–1,000 kcal/day above maintenance requirements in order to gain 1–2 pounds per week.

Treatment plans need to emphasize weight restoration and normalization of eating behaviors. After evaluation of illness severity, available psychosocial supports, prior treatment history, and treatment availability, evaluations should conclude with recommendations for treatment type and setting. Patients, and their families, whenever possible, should participate in treatment planning. Most treatments include a team of clinicians with medical, psychological, and nutritional expertise. Target weight goals (and the rationale for these recommendations) should be provided, along with explicit expectations about rate of weight gain and plans for weight monitoring. Once a recommended weight has been achieved, goals can be reevaluated to incorporate weight maintenance. Psychoeducation about normal fluctuations in weight and healthy weight *ranges* often is required.

Clearly, treatment can be both challenging and complicated, and the lack of clear empirical support for any single treatment approach may be a result of different factors, including several difficulties inherent to treating this clinical group (Attia, 2010). Associated features of AN, such as the ego-syntonic nature of the disorder and the medical complications that can arise during treatment, can have profound implications on patient motivation, compliance, and outcome. Ambivalence about recovery and denial about the seriousness of the problem can be framed as symptomatic of the illness and should be explicitly addressed at the onset of treatment and repeatedly as needed.

Treatment research in individuals with AN is complicated by their varying needs in terms of intensity of services and the tendency for patients to be withdrawn from outpatient treatment for a higher level of care (Bulik, Berkman, Brownley, Sedway, & Lohr, 2007). The rarity of the illness also makes it challenging to recruit adequate samples for rigorous study. Overall, psychotherapeutic interventions for AN target either acute treatment for underweight or the prevention of relapse to a low weight.

Family Therapy

Evidence supports the use of family-based treatment of adolescents with AN, though only a handful of randomized controlled trials (RCTs) have been completed. One type of family-based therapy (FBT), also called the Maudsley approach because of its origins at the Maudsley Hospital in England (Russell, Szmukler, Dare, & Eisler, 1987), stands out from the empirical database. FBT assumes that normal adolescent development has been disturbed by the presence of the eating disorder; thus, parents' involvement in treatment is essential to this therapeutic intervention (Lock & Le Grange, 2012). Parents temporarily oversee the refeeding process, and once successful in this task, gradually return an age-appropriate degree of control and autonomy back to the patient. After eating and weight have normalized, treatment focuses on feelings underlying the disordered behaviors and the developmental issues that may have been ignored due to the illness. This treatment modality is best suited to those under age 18, who live at home with their families, and whose families are able to set aside other issues in the service of treating the eating disorder. The entire household (e.g., parents, siblings, child care providers) is typically involved and must be committed to the approach and the day-to-day mealtime routines.

Overall, this and other family-based treatments are considered the "gold standard" for adolescents with AN (Lebow et al., 2018). Moreover, existing data suggest long-term benefit of FBT (Loeb & Le Grange, 2009). One study of adolescents receiving FBT observed that 78% of patients no longer met criteria for an eating disorder at 36-month follow-up (Paulson-Karlsson, Engstrom, & Nevonen, 2009). Studies comparing subtle variations of FBT suggest that parental involvement may be the most salient feature of the treatment, prompting the development of parent-focused treatment (PFT; Hughes, Sawyer, Loeb, & Le Grange, 2015). RCTs comparing PFT to FBT have consistently found PFT to yield greater reduction in symptoms and increased remission rates (Hughes, Sawyer, Accurso, Singh, & Le Grange, 2019). The evaluating clinician must consider carefully whether parents *and* patients can be adequately committed to this psychotherapy and, accordingly, what interventions to employ.

Cognitive-Behavioral Therapy

CBT, pioneered by Beck (1976), has demonstrated efficacy for a wide range of psychiatric illnesses (Hofmann, Asnaani, Vonk, Sawyer, & Fang, 2012). As previously described, CBT was initially adapted for patients with BN and found to be efficacious (CBT-BN; Fairburn, 1981); it has since evolved into CBT-E.

A central tenet of the CBT model of AN (CBT-AN) is that symptoms are maintained by the interaction between disturbances in thoughts and behaviors that impact eating and weight control. Though available data on CBT-AN remain limited, there is some indication that this approach may be of use preventing relapse in weight-restored patients: Studies comparing CBT-AN to alternative approaches have consistently reported lower rates of premature dropout and longer time to relapse in weight-restored individuals receiving CBT (Agras, 2019; Galsworthy-Francis & Allan, 2014). Despite this, meta-analyses examining CBT-AN suggest its efficacy is not consistently superior to other treatments (Galsworthy-Francis & Allan, 2014).

CBT-AN's efficacy in treating acute AN is somewhat ambiguous. In the first published case series, Cooper and Fairburn (1984) reported that two patients did well, one showed some improvement, and two did not benefit. Although subsequent RCTs have found no differences in outcome for underweight patients receiving CBT-AN compared to other treatments (Galsworthy-Francis & Allan, 2014), a trial comparing CBT-AN with specialist supportive clinical management in a sample of adults with severe and enduring AN found CBT-AN to yield greater social adjustment and symptom reduction at 6- and 12-month follow-up (Touyz et al., 2013). While there is minimal evidence to support this approach over others, existing studies in this subgroup are limited by the methodological problems outlined previously.

Research on CBT-E for AN remains relatively limited, though early reports suggest its suitability for a substantial proportion of patients, with reports of up to 96% of patients successfully completing treatment (Dalle Grave, El Ghoch, Sartirana, & Calugi, 2016). There is some indication that this approach yields increased weight and decreased symptoms among the majority of patients (60–80%) who complete treatment (Dalle Grave et al., 2016) and that gains are relatively well maintained a year after treatment completion (Frostad et al., 2018). Among patients with severe AN (defined as those with BMI < 16 kg/m^2), 50% successfully completed outpatient CBT-E, and 71.4% of these individuals had a substantial increase and maintenance of BMI up to 12 months later (Frostad et al., 2018). The guidelines for eating disorders therefore now recommend CBT-E as a first-line treatment for adults with AN.

Additional Approaches

Given the limited evidence in support of any single psychotherapeutic intervention for AN, clinical researchers have adapted treatments originally studied in other patient populations, such as cognitive remediation therapy (CRT), exposure therapy and response prevention (EX/RP), and acceptance and commitment therapy (ACT) for use in this population.

Developed as an intervention for patients with impaired cognitive functioning due to brain lesions, CRT aims to identify and strengthen impaired cognitive processes through repetitive cognitive and behavioral exercises (Tchanturia & Hambrook, 2010). The therapy addresses areas of relative cognitive weakness, based on neuropsychological assessment, neuroimaging findings, and studies comparing individuals with AN to healthy counterparts. Areas of impairment include "set shifting" (to transition easily back and forth between tasks or mental sets) or "cognitive flexibility" (to adapt behavior to changing demands in the environment), and "central coherence" (holistic information processing whereby details are integrated and put in a broader context). In CRT, patients work on versions of different tasks to address inflexibility in thinking and overly detailed processing biases (for details, see Tchanturia & Hambrook, 2010). CRT appears to yield improvement on neuropsychological measures in patients with AN (Tchanturia, Lloyd, & Lang, 2013) and may reduce treatment attrition rates relative to CBT (Dahlgren & Rø, 2014). Preliminary results from RCTs suggest that when combined with treatment as usual, CRT may enhance certain outcomes, including improvements in eating disorder psychopathology and quality of life (Dahlgren & Rø, 2014).

In EX/RP for AN, disturbances in eating behavior are explicitly targeted. This intervention, predicated on patients' learning through experience that anticipated

negative consequences do not occur (or can be managed) when a fearful stimulus is approached, is the psychotherapy of choice for OCD, posttraumatic stress disorder, and some anxiety disorders (Kaczkurkin & Foa, 2015). In EX/RP for AN, the avoidance of food likely to produce weight gain is hypothesized to result from anxiety. Rigid dietary rules and ritualized eating behaviors (e.g., eating slowly, cutting food in small pieces) are conceptualized as strategies to avoid or minimize distress around eating (Steinglass et al., 2011, 2014). To promote experiential learning, EX/RP for AN utilizes session time to put the patient in direct contact with feared foods and eating situations. In an open case series, Steinglass and colleagues (2012) observed increased caloric intake during a laboratory meal following a brief trial of EX/RP. An RCT of EX/RP and CRT in inpatients showed that EX/RP was associated with better intake in a laboratory meal; furthermore, improvement in intake was associated with improvement in eating-related anxiety (Steinglass et al., 2014).

ACT (Hayes, Strosahl, & Wilson, 2002) emphasizes a need for psychological flexibility and acceptance of the human experience. In this approach, eating disorders are conceptualized as a means of avoiding unpleasant experiences. ACT aims to shift a patient's perspective on how to instead approach and accept these experiences. In case series and larger samples incorporating ACT into existing treatments, improvements in weight and symptom severity are comparable with treatment as usual (Juarascio et al., 2013; Parling, Cernvall, Ramklint, Holmgren, & Ghaderi, 2016). ACT holds promise but is not currently considered superior to other treatments.

Bulimia Nervosa

While individuals with BN are more likely than individuals with AN to present with clear interest in symptom resolution, it is not unusual for them to delay their presentation because of shame. All comprehensive psychological evaluations should include questions about eating, body shape, and weight concerns, and individuals who report concerns in these areas should be asked about binge-eating and purging behaviors. As with AN, assessment should include gathering information about typical overall eating pattern (and associated dietary rules) and binge episodes (e.g., amounts/types of food, context/triggers). Patients with BN may report an escalating pattern of binge and purge frequency. They may even have had an earlier period of restrictive food intake and may have previously met full criteria for AN. Younger patients may report less frequent binge eating and purging. Self-induced vomiting at any frequency should be identified as a problem and addressed in the evaluation and treatment plan. Assessments for BN should include questions about social functioning because patients may eat secretively, or pair restrictive eating when alone with apparently "normal" eating with friends, accompanied by purging following a fuller meal. Patients should be made aware of the risks of frequent vomiting, including dental erosion and decay, and electrolyte instability associated with serious medical risks, such as cardiac arrhythmia or seizures. Individuals who abuse laxatives may develop significant difficulties with intestinal motility, as well as fluid and electrolyte abnormalities.

As symptoms are often the source of great shame, patients should be commended for speaking honestly and seeking help. Patients require education about the cycle of thoughts and behaviors that frequently perpetuate symptoms and need support in

order to break this cycle, introduce healthier eating behaviors, and manage anxiety about whether eating without purging can be accomplished without feared weight gain. Treatment planning should include education about effective treatments for BN, including psychotherapy and antidepressant medication. Regular medical evaluation should be required if patients continue to be notably symptomatic.

There has been great progress in the development and rigorous empirical study of several psychotherapies for BN in the past two decades. This body of research has identified substantial differences in the effectiveness of distinct psychotherapies, allowing for clearly delineated treatment guidelines (Grilo, 2006). In contrast to AN, the prognosis for patients with BN who receive adequate treatment is good. Approximately 50% of individuals with BN who receive manual-based CBT, for example, achieve either near or full remission of symptoms that appears to be maintained over time (Wilson, Grilo, & Vitousek, 2007). Given the robust empirical database of treatment research in BN, in 2017 the U.K.'s National Institute for Health and Care Excellence (NICE) updated their recommended list of evidence-based treatment guidelines, designating CBT-guided self-help (CBT-GSH) as the treatment of choice, followed by CBT-E.

Cognitive-Behavioral Therapy

CBT targets behavioral disturbances by prescribing a regular pattern of eating, providing problem-solving skills to cope with high-risk situations, and teaching cognitive restructuring to modify problematic body shape and weight beliefs (Fairburn, Marcus, et al., 1993).

RCTs of CBT-BN support its efficacy in helping patients achieve abstinence from or a decreased frequency of binge–purge behaviors. Consistent with the approach's conceptual model, the reduction in dietary restraint achieved by CBT partly mediates the intervention's efficacy in eliminating binge eating and purging (Wilson, Fairburn, Agras, Walsh, & Kraemer, 2002). CBT has been demonstrated to be superior to other psychological treatments in the short term and to antidepressant medications for achieving abstinence from bulimic symptoms (Linardon, Wade, de la Piedad Garcia, & Brennan, 2017).

By addressing difficulties external to the eating disorder (as described earlier), CBT-E may be an effective treatment for complex or severe cases, though there does not appear to be a consistent difference between the focused and broad versions (Thompson-Brenner et al., 2016). Studies do provide relatively robust support for CBT-E as an effective treatment, yielding global symptom reduction and significantly improved outcomes when compared with psychoanalytic psychotherapy, interpersonal psychotherapy (IPT), and wait-list control (de Jong et al., 2018). Despite this, CBT-E has relatively high attrition rates, and there is preliminary evidence to suggest alternatives, such as integrative cognitive–affective therapy, may also be effective (de Jong et al., 2018).

GUIDED SELF-HELP USING CBT

Despite the effectiveness of CBT in the treatment of BN, access to this modality of treatment remains limited (Wilson et al., 2007). CBT-GSH was developed to

overcome barriers in treatment dissemination. It combines a self-help manual with limited, brief "therapy" sessions administered by health care providers with differing degrees of expertise and experience.

Interpersonal Psychotherapy

Originally developed as a short-term, structured psychotherapy for depression (Klerman, Weissman, Rounsaville, & Chevron, 1984), IPT has been adapted for BN by Fairburn and colleagues (Fairburn, Jones, Peveler, Hope, & O'Connor, 1993). IPT for BN does not overtly focus on the modification of disturbed eating behaviors or overvaluation of shape and weight. Binge eating is theorized to occur as a response to interpersonal disturbances and consequent negative mood (Fairburn, Jones, et al., 1993). Thus, the primary emphasis of IPT is on helping patients identify and address interpersonal problems hypothesized to maintain the eating disorder. Treatment goals evolve from the four domains of interpersonal problems, as outlined by Klerman and colleagues (1984)—grief, interpersonal role disputes, role transitions, and interpersonal deficits—and encourage mastery of current social roles, as well as adjustment to evolving interpersonal situations (Wilfley et al., 2002). Eating disorder symptoms are linked consistently back to their role in the perpetuation or maintenance of the patient's interpersonal domain of focus (Tanofsky-Kraff et al., 2010).

IPT has some empirical support and therefore might be considered an alternative to CBT. However, RCTs show this approach to be less effective than CBT in reducing bulimic symptoms by the end of the active phase of treatment (Agras, Walsh, Fairburn, Wilson, & Kraemer, 2000; Fairburn et al., 1991) and rates of remission are higher in CBT-E (Fairburn et al., 2015). Nevertheless, studies have shown that individuals receiving either treatment report a similar degree of improvement in bulimic symptoms at follow-up. Thus, there appear to be clear temporal differences in response to treatment, with IPT taking longer to achieve its effects than CBT (Fairburn, Jones, et al., 1993; Wilson et al., 2002). Proponents of IPT assert that if delivered in such a way that interpersonal problems are repeatedly linked to the maintenance of the eating disorder, response to IPT may occur more rapidly (Tanofsky-Kraff et al., 2010).

Additional Approaches

Given the need for psychological treatments to address a broader range of individuals with BN (Wilson et al., 2007), dialectical behavior therapy (DBT) and FBT for BN have garnered increased interest. Some attention has also been paid to leveraging technology to increase access to evidence-based care.

DBT, developed to treat borderline personality disorder (Linehan, 1993), is an empirically supported treatment that focuses primarily on affect regulation and development of adaptive coping strategies and interpersonal skills. The skills taught in DBT, including mindfulness, distress tolerance, and emotional regulation, may be a particularly good match for the high levels of negative affect frequently experienced by patients with BN (Safer, Telch, & Agras, 2001; Wilson et al., 2007). Findings from initial trials suggest that DBT for BN can yield a large reduction of bulimic symptoms. Rigorous studies comparing DBT to an active treatment are needed to better evaluate its efficacy.

The Maudsley form of family therapy described previously has also been adapted to meet the needs of younger patients with BN. Research suggests that FBT is useful in reducing bulimic symptoms (Le Grange, Crosby, Rathouz, & Leventhal, 2007). While previous studies suggested FBT may not be more effective than other approaches, more recent findings indicate that when compared to CBT for adolescents, FBT for BN results in greater abstinence from binge eating and purging at end of treatment and 6-month follow up (Le Grange, Lock, Agras, Bryson, & Jo, 2015).

Beyond meeting the needs of a broader range of patients with BN, there has been a growing interest in telemedicine as a strategy to reach more individuals and provide cost-effective treatment. Internet-based CBT has shown promise for individuals with anxiety disorders (Andersson, 2009). Some data suggest telemedicine CBT-BN yields outcomes greater than control conditions and is comparable to face-to-face CBT, but findings are mixed (Loucas et al., 2014; Pittock, Hodges, & Lawrie, 2018). More research is needed to address logistical considerations in delivering telemedicine, including the use of smartphone apps, and to identify possible long-term differences in outcome between this and other treatment modalities.

Binge-Eating Disorder

BED is the most prevalent eating disorder and occurs in many individuals who present for treatment for other psychiatric disorders, and in some who present at obesity clinics and other medical venues. Therefore, individuals with mood disorders, psychotic disorders, anxiety disorders, and obesity should be assessed for binge eating. Treatment planning for BED requires patient education and participation. It is not synonymous with weight loss treatment, and treatments that effectively decrease bingeing may not be associated with weight loss. Psychotherapy for BED is often one of several treatment components that overweight or obese individuals with BED may want to consider in moving toward improved eating and health.

Multiple therapeutic modalities show promise. Initial treatment studies came from the field of obesity, where mixed findings led to concern that behavioral weight loss interventions and calorically restrictive diets might be unsuitable for obese individuals endorsing features of eating disorders (Yanovski, 1993). For the many obese individuals with BED, there continues to be debate in the eating disorder and obesity fields as to whether weight status or eating disorder symptoms should be first-line treatment goals (Krysanski & Ferraro, 2008). Given limited data supporting maintenance of significant weight loss following behavioral weight loss, targeting a reduction in binge eating initially may allow for a slow, modest weight loss following BED treatment (Krysanski & Ferraro, 2008). However, research including long-term follow-up is limited. Additionally, more studies comparing obese and nonobese individuals with BED are warranted to clarify the extent to which the eating disorder diagnosis offers clinically relevant information in relation to obesity (Wonderlich et al., 2009).

The clinical concerns of practitioners treating obese individuals with BED inspired research on psychological treatments adapted from those for BN. Currently, CBT-GSH enjoys the designation as the treatment of choice for BED by the U.K.'s NICE. In considering the treatment needs of individuals with BED, however, it is important to bear in mind that these patients struggle across domains (e.g., binge eating, attitudinal features of eating disorders, general psychological symptoms and distress, and, at times, obesity with its associated health risks; Grilo, 2006).

Cognitive-Behavioral Therapy

Treatment studies evaluating CBT for BED (CBT-BED) have typically relied upon the manual developed by Fairburn, Marcus, and colleagues (1993) for the treatment of BN. Patients are encouraged to challenge negative cognitions related to having a larger body size and to develop structured, healthy eating patterns, with an emphasis on moderation of amounts and types of foods. Adjustments to the theoretical understanding of maintenance mechanisms recognize key differences between the disorders, namely, the absence of inappropriate compensatory behaviors and more generally chaotic, less restrictive eating characteristic of BED (Grilo, 2006).

In RCTs of CBT-BED, substantial reductions in binge eating, and general and eating-disordered psychopathology are evident posttreatment and at follow-up (Wilfley et al., 2002). Though CBT interventions do not reliably result in substantial or sustained weight loss, they yield greater reductions in binge eating than behavioral weight loss (Iacovino, Gredysa, Altman, & Wilfley, 2012). CBT is generally associated with good treatment completion rates, reduction in binge-eating episodes in more than half of patients, and improvement in psychosocial functioning and mood. CBT has proven more effective than pharmacological treatments as a stand-alone treatment and as an adjunct to behavioral weight loss treatment (Iacovino et al., 2012).

While data clearly and consistently support its use, little is known about how CBT-BED produces its effects. Some have proposed that the current CBT model pays insufficient attention to low levels of dietary restraint, high negative affect, and eating in response to emotion characteristic of individuals with this eating disorder (Wilson et al., 2007). A review of recent research on CBT-E provides evidence that this is also an effective treatment for BED and results in faster treatment response and higher remission rates than IPT (de Jong et al., 2018).

GUIDED SELF-HELP USING CBT

Similar to what we described for BN, self-help based on the CBT model of eating disorders has been studied extensively, primarily using the Fairburn (1995) manual. Several investigations have tested CBT-GSH and pure self-help approaches (for a complete list, see Sysko & Walsh, 2008); in BED, data suggest some advantage of CBT-GSH over pure self-help (Iacovino et al., 2012). In one study, 10 sessions of CBT-GSH delivered by beginning psychology graduate students with limited therapy experience proved to be as effective in reducing BED symptoms as 20 sessions of IPT conducted by intensively trained and supervised doctoral-level psychologists (Wilson, Wilfley, Agras, & Bryson, 2010), providing further support that CBT-GSH may be a viable, appropriate intervention with advantages in dissemination and cost-effectiveness.

Interpersonal Psychotherapy

IPT for BED (Wilfley et al., 2002) is based on IPT for depression (Klerman et al., 1984) and BN (Fairburn, Jones, et al., 1993). Although connections are made throughout therapy between interpersonal events and binge eating, the treatment does not overtly address eating behavior. Instead, patients' work focuses on making changes in specifically identified problematic areas of interpersonal functioning. In adapting IPT for BED, Wilfley and colleagues (2002) developed a group format. In group IPT,

patients are encouraged to use the group to decrease isolation, enable the formation of relationships, and practice new communication skills. The group format provides a unique environment for participants, who may otherwise keep eating behaviors hidden due to guilt and shame (Tanofsky-Kraff et al., 2010).

At present, NICE (2017) guidelines for BED do not designate IPT as a treatment of choice, possibly because of the paucity of RCTs investigating this approach relative to those of CBT-based treatments. Still, research underscores the potential utility of an IPT approach in a subgroup of patients known to suffer across domains of functioning (Grilo, 2006). In the largest RCT to date, IPT was found to be superior to behavioral weight loss in patients with low self-esteem at 2-year follow-up, and superior to CBT-GSH in patients with low self-esteem and high eating disorder psychopathology (Wilson et al., 2010). While this suggests that IPT holds promise as a treatment for a subset of individuals with BED, it remains secondary to CBT-based approaches.

Additional Approaches

Over the past several years, ACT's theoretical basis has been used to inform novel treatments for BED. For example, an acceptance-based behavioral treatment (ABBT) for BED, designed to combine acceptance-based principles from ACT and DBT with behavioral strategies from CBT, has been found effective in reducing global eating disorder symptoms, including binge-eating frequency (Juarascio, Manasse, Schumacher, Espel, & Forman, 2017). Despite this, the reported binge-eating abstinence rates following this treatment suggest it is comparable to CBT (Pisetsky, Schaefer, Wonderlich, & Peterson, 2019). In a study of the feasibility and acceptability of ABBT combined with behavioral weight loss to target both binge eating and weight loss in overweight and obese adults with BED, patients were able to concurrently achieve minimal weight loss while managing eating disorder symptoms (Evans, Murray, Muratore, Lantz, & Juarascio, 2019). ACT may be a suitable treatment for BED, but it requires additional investigation.

Another treatment approach with preliminary support is DBT (Linehan, 1993). DBT has led to a greater reduction of binge-eating frequency and decreased eating-disordered psychopathology compared to no treatment, and has been associated with greater initial efficacy and fewer treatment dropouts when compared with an active comparison group therapy (Iacovino et al., 2012; Wilson, 2011). Although symptom reduction appears to be maintained at follow-up (Wilson et al., 2010), findings suggest DBT is not superior to CBT or CBT-GSH (Chen et al., 2017) and, when delivered in routine practice settings, may be associated with increased attrition rates (Klein, Skinner, & Hawley, 2012).

Avoidant/Restrictive Food Intake Disorder

Unlike individuals with other types of eating disorders, those with ARFID experience a problematic eating pattern that is not motivated by body shape or weight concerns. Concerns about appearance and weight may be absent or within a normal range. Differential diagnoses of ARFID and AN can be tricky because the clinical presentation of both disorders can involve low weight, and individuals with AN

may deny or minimize fear of fat, or body image or weight concerns. However, those with ARFID who are underweight sometimes describe feeling unhappy about being thin and endorse feeling proud of or happy about weight gain during treatment. Moreover, the foods preferred by individuals with ARFID tend to be energy-dense, high-fat, and high-carbohydrate. Food restriction or avoidance is often guided by principles related to color or texture, or due to lack of interest in food and eating, rather than macronutrient content. Compared to individuals with AN, those with ARFID consume a daily diet with a significantly lower ratio of protein (Thomas & Eddy, 2018). Weight trajectories over time offer another possible distinction between ARFID and AN. Those with low-weight ARFID tend to have been at chronically low weight prior to seeking treatment, rather than describing an acute period of weight loss (Strandjord, Sieke, Richmond, & Rome, 2015).

Assessment of ARFID includes careful detailing about typical eating patterns, with an emphasis on clarifying which foods are overly relied on or avoided. Collateral information from parents or other individuals close to the patient is helpful. Evaluation should also include inquiries about the rationale for food restriction or avoidance, with questions about sensory sensitivity, fear of aversive consequences, disinterest in eating, and psychosocial impairment. Finally, assessment of the medical sequelae of restrictive eating is essential. This involves measurement of height and weight, consideration of the growth trajectory in children and adolescents, and assessment of vitamin deficiencies and use of supplements.

The current evidence base for ARFID treatment primarily comprises case reports, case series, chart reviews, and a few controlled trials in young children. Hospital-based refeeding programs appear to be helpful for children and adolescents with low-weight ARFID and medical instability (Thomas, Wons, & Eddy, 2018). Outpatient psychological treatments are currently being studied, but results have not yet been published.

Cognitive-Behavioral Therapy

Common components of CBT for ARFID include regular eating, self-monitoring of food intake, exposure (to new foods and feared eating-related situations) and response prevention, relaxation training, and behavioral experiments (Thomas et al., 2018). The proposed disorder-specific CBT model hypothesizes that some individuals have a biological predisposition to sensory sensitivity (e.g., heightened response to unfamiliar tastes and smells), fear of aversive consequences such as choking or vomiting (e.g., high trait anxiety), and/or lack of interest in food or eating (e.g., low hedonic appetite). Individuals with this predisposition may be particularly apt to developing negative feelings about food and assumptions about eating that lead to the restriction or avoidance of foods. For example, someone with ARFID may think, "Whenever I try a bite of an apple, I gag because it tastes so terrible, so I will probably not like any fruit," and over time repeatedly rely on items from any other food group, limiting opportunities for exposure and putting them at risk for compromised nutritional status.

CBT for ARFID (CBT-AR; Thomas & Eddy, 2018) is a 20- to 30-session intervention being evaluated in individuals age 10 and up (through adulthood) that aims to (1) provide education, (2) normalize intake by increasing food volume and variety,

and (3) address variants of avoidant/restrictive eating using systematic desensitization to novel foods, guided exposures to feared eating situations, and/or interoceptive exposure to bloating, fullness or nausea.

Family-Based Therapy

FBT for ARFID is similar to FBT for AN, in that parents are empowered to take charge of feeding. However, it differs in that parents are additionally explicitly tasked with helping their children increase dietary variety with repeated exposure to novel foods and, depending on the subtype of ARFID, involvement of other family members may not be required (Lock et al., 2019). In addition, a key focus in FBT for ARFID is helping caregivers to counter what may be long-standing accommodations that have been made to disordered eating behaviors.

MEDICATION USE

Anorexia Nervosa

Currently, there is no clearly established role for medication use in the treatment of acute AN. Given the prevalence of AN symptoms that are commonly responsive to medication in other clinical populations (e.g., depression, anxiety, and obsessionality), several classes of medication (e.g., antidepressants, anxiolytics, appetite-enhancing drugs) have been examined in patients with AN in small RCTs (Hay & Claudino, 2012), without effect on weight or psychological symptoms. Investigators have theorized that the lack of treatment response across these medication classes may result from physiological changes that accompany malnutrition. Generally, psychological symptoms improve significantly with weight restoration alone (Meehan, Loeb, Roberto, & Attia, 2006). Recently, however, a large, multisite study comparing the atypical antipsychotic medication olanzapine to placebo found that this medication assists in achieving modest weight gain for outpatients (Attia et al., 2019), confirming reports from prior smaller studies (Attia et al., 2011; Bissada, Tasca, Barber, & Bradwejn, 2008).

Several studies have also considered the utility of medication for relapse prevention in weight-restored patients. Despite initial reports that fluoxetine might be helpful (Kaye et al., 2001), a large RCT observed fluoxetine to be no more beneficial than placebo in preventing relapse during the first year after hospitalization among patients receiving CBT (Walsh, Sysko, & Parides, 2006). However, once patients have achieved and maintained normal weight for some time, a medication evaluation may be warranted to assess for persistent co-occurring symptoms.

Bulimia Nervosa

Medication treatment for BN should be considered when an individual prefers a medication approach, in the case of poor or partial response to CBT, or when CBT is unavailable. Generally, medication trials have demonstrated that a variety of medication classes are superior to placebo in reducing binge–purge behavior (Bacaltchuk & Hay, 2003; Shapiro et al., 2007). Fluoxetine is commonly tried first, as it is the

only medication with a specific indication for the treatment of BN by the U.S. Food and Drug Administration (FDA; Broft, Berner, & Walsh, 2010). Psychotherapists are advised to collaborate with prescribing physicians to monitor bulimic symptoms if patients are using or are deciding to come off medication as part of treatment. Patients should be monitored for symptom relapse after discontinuing medication.

Binge-Eating Disorder

Most, but not all, controlled medication studies for BED have observed a significant advantage for medication versus placebo in the short-term reduction of binge eating (Reas & Grilo, 2008). Investigations have largely evaluated the efficacy of different antidepressants, an anticonvulsant (topiramate), and weight loss medications including orlistat and sibutramine. Recently, the stimulant lisdexamfetamine has been identified as superior to placebo in both its effect on binge frequency and weight change (McElroy et al., 2017), leading to FDA approval for its use in BED and representing the only medication that has received this indication. There does not appear to be substantial difference in the efficacy of these different agents relative to placebo in treating BED psychopathology (Grilo, 2006). Fluoxetine has also been associated with improvement in depressive symptoms in the short term (Devlin et al., 2005) and at follow-up (Devlin, Goldfein, Petkova, Liu, & Walsh, 2007) for individuals with BED. Not surprisingly, the use of obesity medication has been observed to result in more clinically meaningful weight loss concomitant with a reduction in binge frequency (Wilfley et al., 2008). A comprehensive meta-analysis of BED pharmacotherapy studies also determined that orlistat and topiramate offer enhanced weight loss as adjunctive treatment to CBT and behavioral weight loss (Reas & Grilo, 2008). Thus, medication might be recommended as part of a multidisciplinary treatment approach for BED, particularly for those who are significantly overweight or suffering from additional psychological problems.

TREATMENT MOTIVATION

The issue of motivation for treatment is embedded in the previous discussion of treatment approaches. As described, individuals with AN most frequently do not self-present for treatment, and the ego-syntonic nature of the disorder is perhaps its most dramatic and particularly challenging clinical feature (comparable to individuals with substance abuse disorders). Because investment in an excessively low weight is core to AN, treatment interventions must address motivational issues from the start. One of the particularly salient components of the FBT approach is to engage parents to assist with and lead in the recovery process, until their child is in a healthier state, such that the child is aligned with and able to assume responsibility for recovery. In the case of BN and BED, goals regarding weight loss may compete with recovery from the eating disorder. For ARFID, fear of or disinterest in particular foods can impact the stamina required for consistent, repeated exposure trials. Long-term success of eating disorders treatment ultimately depends on the patient's sense of ownership about the treatment goals and commitment to recovery goals. Clinicians' creativity (e.g., finding meaningful metaphors as patients grapple with

ambivalence) and flexibility (e.g., developing and operationalizing mutually agreed-upon intermediate goals while working toward long-term goals) may enhance therapeutic alliance and motivation for change.

THE PROCESS OF CHANGE IN EVIDENCE-SUPPORTED TREATMENTS

There is increasing interest in the field of psychotherapy research in exploring treatment mechanisms and identifying factors that may contribute to behavioral change in the context of successful treatment. While Castonguay and Beutler (2005) have outlined participant, technique, and clinical relationship factors that should be considered in examining principles of therapeutic change associated with evidence supported treatments, very little has been written about such principles associated with treatments for eating disorders. Treatment elements relevant to CBT that may represent principles for change include the challenge of cognitive distortions about body shape and weight, as well as the reinforcement of structured, healthy eating behaviors. Elements relevant to IPT and possibly reflective of principles of change associated with this approach include encouragement of normal socialization, cessation of social isolation, and development of communication skills to improve relationships and decrease eating in response to emotion. Elements relevant to FBT for AN and ARFID include structured eating and changing the environment to make eating and weight gain more likely.

Several investigators have examined variables that may predict treatment outcome or serve as mediators for change. Across studies of different treatments, early treatment response appears to predict both shorter- and longer-term outcomes. Specifically, improvement in binge eating and purging during the first 4 weeks of CBT or IPT treatment (Agras et al., 2000), as well as during the first few weeks of antidepressant treatment (Walsh et al., 2006), predicts symptom remission at the end of treatment for individuals with BN. While it is unclear whether there is something inherently different (and more likely to remit) about the individuals who respond quickly to treatment, these findings suggest that treatment should emphasize behavioral change and that individuals who do not experience symptom resolution early in a treatment course may need a change in the intervention. Relatedly, in an analysis of variables thought possibly to mediate treatment response, Wilson and colleagues (2002) found that improvement in dietary restraint mediated improvement in binge eating in patients with BN receiving either CBT or IPT in a randomized trial. This finding was surprising, as normalization of eating behaviors was a direct treatment focus in CBT and not IPT, yet normalization of restraint appeared to be a mediator of both therapeutic approaches, questioning whether the mechanisms of therapeutic effect for these two seemingly different treatments are really so different (Wilson et al., 2002). Among weight-restored individuals with AN, consumption of a diet higher in variety and caloric density predicts likelihood of maintaining normal weight during the year following acute weight restoration. Taken together, these findings on predictors and mediators of outcome indirectly suggest that normalizing eating behavior is central to better outcomes in eating disorder treatments.

TREATMENT INTENSITY

Available services for treating patients with eating disorders range from varying levels of outpatient care (e.g., weekly or biweekly psychotherapy, nutritional counseling, psychiatric care) to intensive outpatient, residential, and inpatient programs. Because specialized programs are not available in all geographic regions, it can be difficult to access appropriate care. Nonetheless, we present guidelines below for how clinicians might choose the most appropriate level of care based on patients' clinical status. Relevant available data on treatment intensity are included.

Anorexia Nervosa

While the specific mechanism responsible for psychological changes in individuals with AN remains uncertain, the improved nutritional state that accompanies refeeding is presumed to enable better cognitive functioning and emotional regulation (Attia & Walsh, 2009). Therefore, weight restoration is an essential primary treatment goal. However, because patients with acute AN have considerable difficulty gaining weight outside of a highly structured treatment, many of those who seek treatment are ultimately guided to an inpatient or residential treatment program (Guarda, 2008).

Admission to an inpatient unit is recommended for adult patients who have lost a substantial amount of weight, particularly if the weight loss has been rapid, and for those whose weight is extremely low for their height (e.g., <75% ideal body weight; Attia & Walsh, 2009). Inpatient treatment may also be appropriate for youth with AN who do not gain the expected amount of weight for their height and age (Attia & Walsh, 2009). Some clinicians may be reluctant to recommend inpatient treatment, but data suggest that outcome is better if inpatient treatment is pursued *before* the onset of medical instability secondary to low weight (Yager, 2007).

The limited availability of specialized inpatient units that treat eating disorders, together with the significant expense of hospital treatments and challenges of third-party reimbursement, often make it difficult for patients to find appropriate hospital programs or to stay in hospital programs for as long as may be recommended. Changes in U.S. health care economics have dramatically shortened hospital stays, and the limited weight gain achieved by patients with AN on inpatient units raises the question of whether shorter stays and lower discharge weights may contribute to higher relapse rates (Wiseman, Sunday, Klapper, Harris, & Halmi, 2001).

As hospital programs have become less available, newly developed residential and structured behavioral day treatment programs have filled the gap. To date, clinical outcome information from residential treatment programs is limited, and while many programs provide some data on their websites as part of promotional marketing efforts, this information has not typically been through a rigorous peer review process (Attia, Blackwood, Guarda, Marcus, & Rothman, 2016; Guarda, Wonderlich, Kaye, & Attia, 2018). Structured day treatment programs have demonstrated some efficacy in helping patients to normalize weight, albeit at a more modest pace. One uncontrolled comparison of specialized inpatient versus day treatment behavioral programs observed an average weight gain in patients of approximately 4 pounds

per week and 2 pounds per week, respectively (Guarda, 2008). Overall, outcomes from specialized day treatment programs are strongly associated with treatment intensity. More successful programs require treatment participation at least 5 days per week, 8 hours per day (Olmsted, Kaplan, & Rockert, 2003). While intensive, structured programming helps in the short-term management of AN, the long-term benefit remains unknown (Attia & Walsh, 2009).

Outpatient treatment for acute AN may be considered for adolescents whose families are committed to FBT aimed at weight restoration (Lock & Le Grange, 2012). Otherwise, outpatient intervention is typically indicated for the prevention of relapse following weight restoration in a structured setting or for patients with symptoms of AN that are at borderline diagnostic threshold (e.g., individuals at approximately 85% ideal body weight with associated features of AN). Weekly psychotherapy is an opportunity for patients to normalize their attitudes about weight, body shape, and eating behavior. Outpatient appointments also provide an opportunity for clinicians to monitor the weight status. If patients are unable to achieve or sustain a healthy weight range, clinicians should recommend an increase in session frequency to twice per week, adjunctive nutritional counseling sessions, or a structured behavioral program.

Bulimia Nervosa

Many patients with uncomplicated BN do not require hospitalization. Depending on the frequency of binge–purge behaviors and severity of associated symptoms, clinicians should consider biweekly psychotherapy appointments. In a standard course of CBT-BN, for example, it is recommended to start with two scheduled sessions per week for the first month of treatment to help patients implement self-monitoring techniques, adopt a regular eating pattern, and break the binge–purge cycle (Fairburn, Marcus, et al., 1993). In initial biweekly IPT sessions, the clinician completes a detailed analysis of the interpersonal context in which the eating disorder has been developed and maintained (Agras et al., 2000). After the early phase of treatment, with some symptom remission, psychotherapy sessions may occur weekly, then bimonthly.

If a patient with BN does not respond to an adequate trial of outpatient psychotherapy, a higher level of care should be considered. Additional indications for more structured treatment include severely impairing symptoms that have not responded to adequate trials of psychotropic medication, serious concurrent medical problems, or psychiatric conditions requiring an inpatient treatment independent of the eating disorder (Yager, 2007).

Though inpatient treatment has long been recognized as *the* viable alternative to outpatient care for treatment-resistant BN, day treatment is an alternative along the continuum of care. Day treatment programs offer a cost-effective alternative to receive care in a more structured setting, while maintaining contact with the natural environment (Zipfel et al., 2002). In one RCT of inpatient treatment versus day treatment for patients with severe, chronic BN, inpatient and day treatment were observed to be equally effective in reducing bulimic symptoms and general psychopathology at discharge and 3-month follow-up (Zeeck et al., 2009). By 12-month follow-up, day treatment demonstrated an advantage in sustained improvement in

binge frequency and bulimic cognitions, but no differences in purging behavior or general psychopathology. This study offers some evidence that day treatment may provide patients with the skills needed for alleviation of symptoms and the opportunity to apply and possibly generalize skills at home.

Binge-Eating Disorder

Individuals with BED who seek treatment typically express several treatment goals, including the cessation of binge eating, the development of healthy eating patterns, and (frequently) weight loss (Devlin et al., 2007). While there is growing evidence that those with BED show greater amelioration of binge eating and associated psychopathology with specialty eating disorder treatment rather than nonspecific weight loss interventions (Wonderlich et al., 2009), there are no data regarding the relative utility of more versus less intensive treatment for this group.

Avoidant/Restrictive Food Intake Disorder

Individuals with ARFID commonly present to medical providers and are generally referred to eating disorder specialists only when low weight, nutritional deficiency, or serious psychosocial disturbance is present. As there is less of an evidence base for treatment, treatment approaches and settings for ARFID are usually those considered for other eating disorders, especially those developed for low-weight AN. For example, individuals with ARFID who need medical stabilization and weight restoration are often referred to inpatient or residential programs. Less medically acute cases are usually managed as outpatients by a team of clinicians.

CONCLUSION

Although the knowledge base regarding prevalence and treatment for eating disorders is continually evolving, sufficient research exists to guide treatment at many points of care for individuals with AN, BN, and BED. Treatment research for ARFID is nascent, but clinical trials are under way. Eating disorders clinicians must rely on professional experience and practice when data fail to inform treatment decisions. One of the challenges for the field is harnessing this clinical experience in a way that furthers an empirically informed approach to care. The complementary challenge is dissemination of study findings, so that clinicians in practice remain up to date on recommendations derived from the empirical research. Relatedly, expanding training opportunities in evidence-based treatments will greatly enhance our capacity to assist individuals with eating disorders most effectively in their journey to recovery.

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